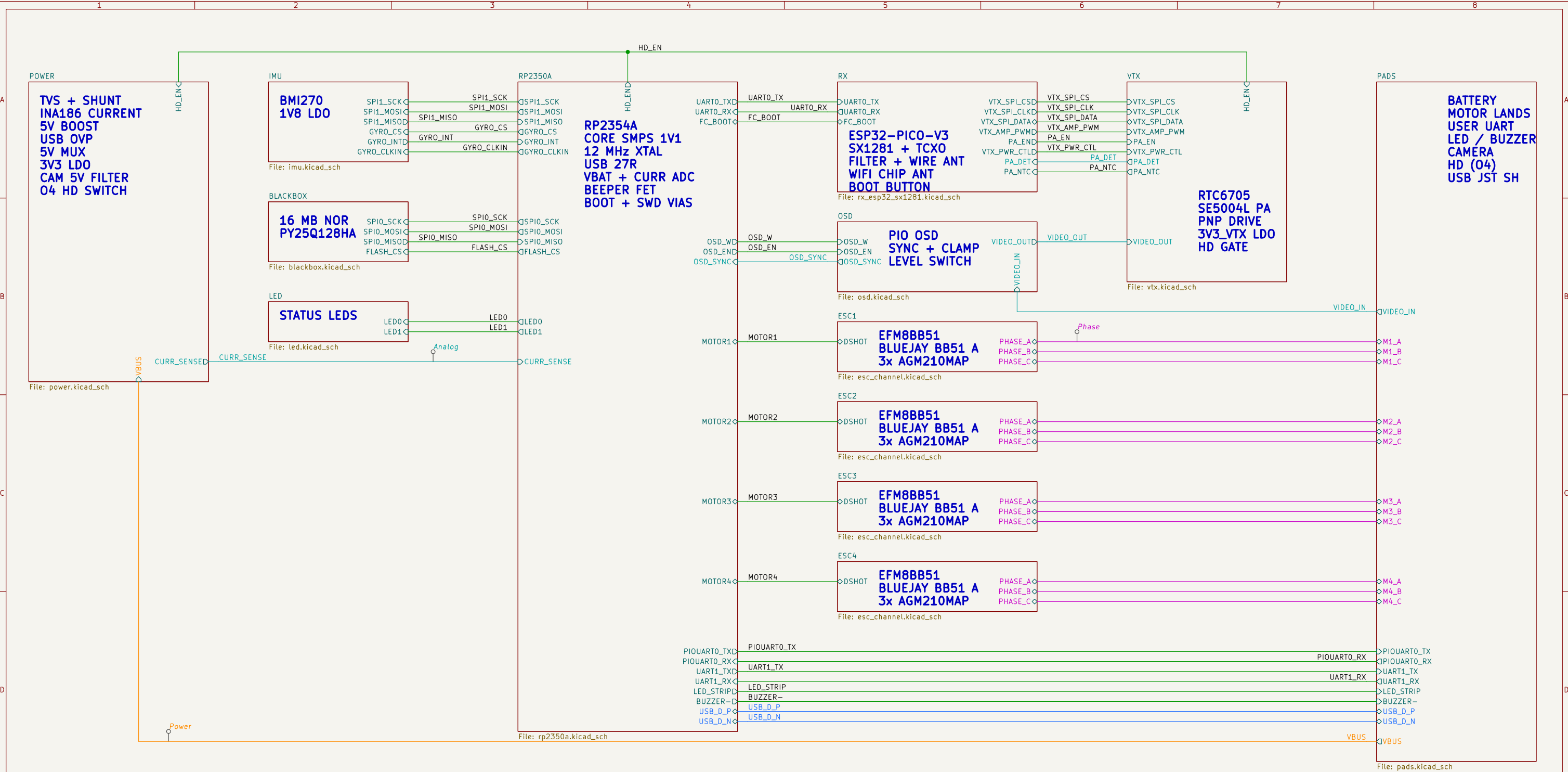




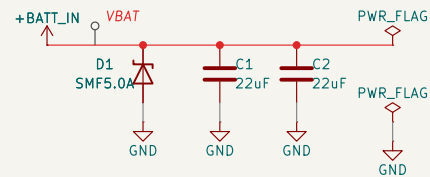
POWER TABLE

Rail	Source	Nominal	Budget (spec 5)	Feeds
+BATT_IN	battery pads J2 / J3 (1S)	3.0–4.35 V	4x ESC + 2.0 A board	TVS, pad bulk, shunt R1
+BATT	shunt R1 0.5 m0hm	cell	ESC + boost in 1.4–3.0 A	ESC1–4, boost, 04 EN, VBAT ADC, PA_VCC
/VBUS	USB J31 pin 4 (hier. net)	5 V	0.42 A, USB only	U24 OVP
+5V_USB	U24 SGM40661 OVP	5 V	as VBUS	U3 VIN1, PR1 divider
+5V_BST	U2 TPS61022 boost	5.15 V	0.70 A analog / 1.48 A HD (BEC 1.5 A cont.)	U3 VIN2, U5 04 switch
+5V	U3 TPS2116 mux	5 V	0.70 A analog, about 0.25 A HD	3V3 LDOs, CAM filter, 5V pads
+5V_CAM	FB1 ferrite from +5V	5 V	0.12 A	camera pad + plug
+5V_HD	U5 TPS22810 (HD_EN, cell)	5 V	1.20 A 04 Lite	VHD pad
+3V3	U4 LP5912–3.3	3.3 V	0.20 A (0.48 A Wi–Fi bench)	FC, IMU, NOR, OSD, RX, LEDs, INA186
+3V3_VTX	U21 LP5912–3.3 (off in HD)	3.3 V	0.10 A typ, about 0.17 A at max PA	RTC6705, PA drive, PA ref
+1V8	U12 TPS7A2018	1.8 V	< 1 mA	BMI270
+1V1	RP2354A core SMPS	1.1 V	internal	RP2354A DVDD
PA_VCC	R67 selector, +BATT default	cell	0.65 A at 27 dBm	SE5004L PA (local, VTX)
PA_VREF	U22 LP5907–2.85 (EN = PA_EN, off in HD)	2.85 V	about 10 mA	SE5004L VREF/EN (local, VTX)
GND	battery –	0 V	–	all sheets

Generated from hardware/sch_contract.json by hardware/tools/sch_root.py; do not edit by hand.
Root wires carry the project-wide net names (root label = net /NAME). Motor phases are named by the ESC sheets (/ESCn/PHASE_x). Rails are power symbols (table); VBUS is a hierarchical net.
ESC1–ESC4 drive M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad–X); FC pin map: research/PINMAP.md.

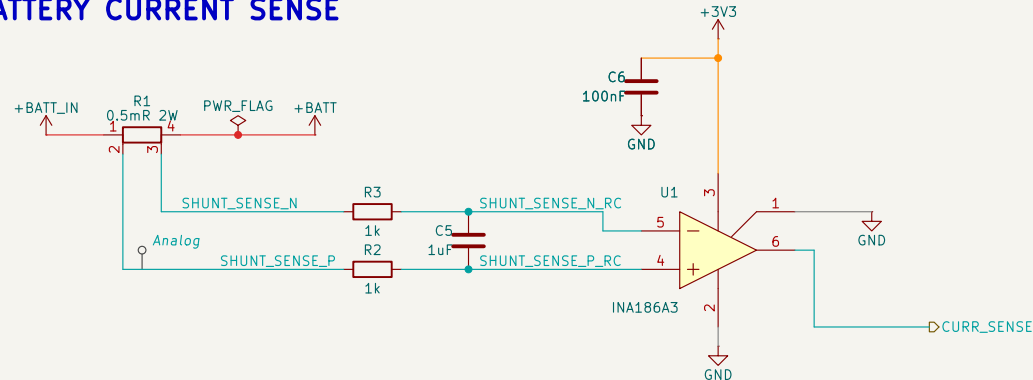
OpenDrone

BATTERY INPUT AND TVS



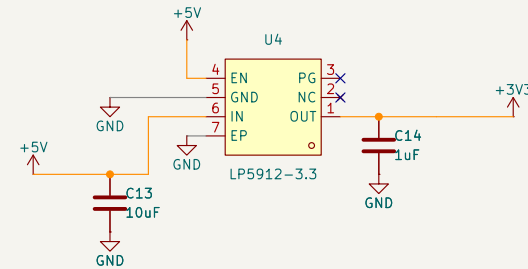
Source: OpenAIO/hardware/OpenAIO.kicad.sch (battery entry), adapted.
+BAT1JN = battery pads J2 (+ / -) J3 (-) on the PADS sheet, 1S LiPo/LiHV
3.0-4.35 V; PWR_FLAG here (the pads are passive). No reverse-polarity FET (spec
4.1: user-fitted pigtail, risk in README).
D1 SMF5.0A: VBR 6.40-7.00 V (10 mA). VC 9.2 V at 21.7 A, 200 W 10/1000 us:
takes one motor's regen energy (-0.29 J per rated pulse) on a battery disconnect:
fallback SMAJ5.0A (ESM 400 W) if V3b exceeds it. It does not hold the 5.5-7 V
parts in rating (FSM4 VDD on the cell; TPS61022 VIN 7 V abs max, and its
ps-throwing MOSFETs are tied to +5V_BST = TPS2116 VIN2, 6 V abs max):
Matrix-parity risk spec. 15.4.
C1/C2 22 uF 16 V X5R 0603 at the pads: 9-13 uF effective each at 4.35 V. Hot
plug of a 4.35 V pack can overshoot to 5.0-5.4 V (V3C): above the TPS61022 4.8
V VIN limit for a start-up with VOUT pre-biased to -0.7 V (DS 6.3), which is
exactly hot plug (EN = VIN, +5V_BST at 0 V); V3C scopes VIN with the +5V_BST
start. Damping fallback -100 uF polymer at the pads: PE-8 waits for V3C.

BATTERY CURRENT SENSE



Source: OpenAIO/hardware/OpenAIO.kicad_sch (shunt + INA186A3, root sheet); input RC moved to 0201 and its two common-mode caps deleted (DS2C BR-07); the output RC (1k/100n) sits at the ADC pin (RP2350A sheet, R47/C60).
P4-9: R1 HC5206 0.5 mOhm on the four-pad Kelvin net-tie land (pad groups 1-2 and 3-4). SHUNT_SENSE_PC = pad 2 (battery side).
SHUNT_SENSE_N = pad 3: each holds only its inner sense pad and R2/R3; SHUNT_SENSE_PC/N_RC hold R2/R3, C5 and U1 (netlist checked); all four match the Analog and Kelvin DRU patterns. Route the pair on L1 or L6, never in2/In3 (DRU). class Analog. Everything that draws current taps the R4.
R2/R3 1k + C5 1 uF differential: $c = 1 / (2 \pi \times f \times 20 \text{ kOhm} \times 1 \text{ uF}) = 80 \text{ Hz}$, $\text{ILB} < c = 3 \text{ nA} \times 1 \text{ kOhm} = 3 \text{ uV}$ (6 mA), $\text{VOS} < 50 \text{ uV}$ (0.1 A).
U1 INA186A3, gain 100: 0.5 mOhm $\times 100 = 50 \text{ mV/A}$, Betaflight ibata_scale 500: output to VS - 40 mV gives about 65 A full scale on +3V3; VCM -0.2..0.4 V covers the cell; REF to GND (unidirectional); C6 100 nF at VS. Shunt loss 0.2 W at 20 A. Scale checked on every prototype (V-pin).

3.3 V LDO FOR FC + RX



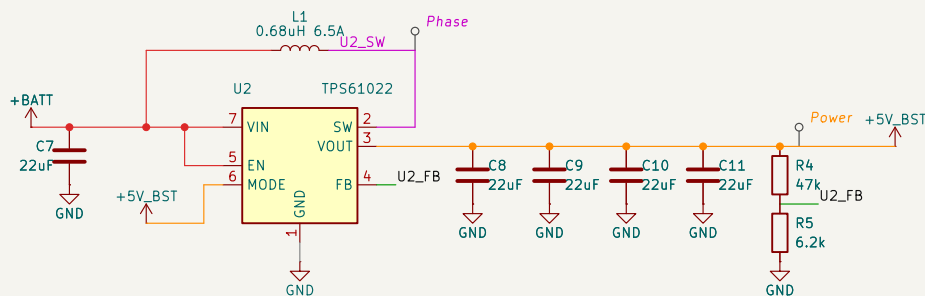
Source: OpenFC-Lite-Mini/hardware/power/kicad.sch (3.3V 0.5A MAIN LDO", LP5912-3.3), with the OpenDrone LP5912-3.3DRVR symbol (pins in package order: IN/EN on the left here, EN tied to +5V, no EN shared with another rail; U21 on VTX has its own EN, PG and NC open; EP to GND for heat (T1: GND or open)).

P4-17 / BR-14: C13 10 uF X6S 0402 is the one CIN of U4 and U21 (VTX sheet): place it between the two IN pins (-2.4 mm apart). 2.0 uF effective at 1.5 V (Murata curve) > the 0.7 uF LP5912 minimum.

C20 1.2-3V3 0.001 at 3.3 V (C82 10 uF 5.0 uF, Murata CWS5C80J106M11 DC-bias curve). C58 4.7 uF 0402 X5R -2.0 V, 3x 1 uF 0201 X5R (C14, C61, C92) 10 uF each, 150 uF of X6S 0201 -0.085 uF each (last three typical, i.); 8.7 uF typical, 7.1 uF at -2.0 V, 10.3 uF only with every part at +20 % -> Inside T1: 0.7-10 uF (SNSV470 7.6) but for that +3 % corner, and most of it is remote (5-30 mm of trace), which the LP5912 allows. Lever if a V-bench load step rings: PS7-06 (delete C92). +3V3_VTX is summed on the VTX sheet.

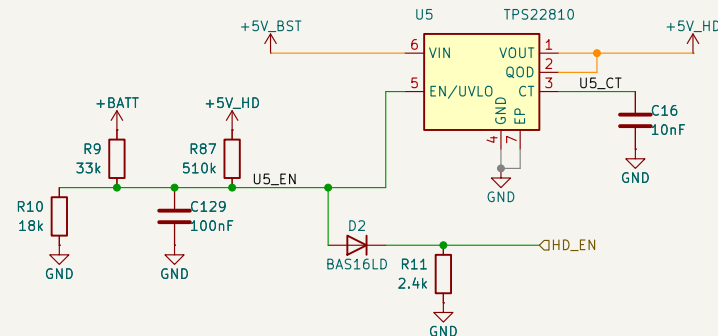
Load 0.20 A: (5.15 - 3.3) V x 0.2 A = 0.37 W, +26 K (RthJA 71.2 K/W). U4 UOT (project-lib LP5912 copy, power=not pin) drives +3V3: no PWR_FLAG.

5 V BOOST TPS61022



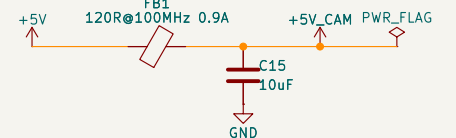
New block from TI: SLVSDXD7D (naming using the TPS61022); look matched to the OpenFxC-Lite-Mini buck blocks.
P4-7: VOUT = 0.600 V $\times$ (1 + 47k / 6.2k) = 5.148 V, with VFB 585=615 mV and 1 $\times$ resistors 4.93=5.37 V. R5 6.2k $\leq$ 300 kOhm (TI). EN tied to VIN (+BATT) and nothing else (BR-11); same -0.3..7 V abs max, VEN_H $\leq$ 1.2 V, on whenever the cell is above UVLO. MODE = VOUT (+5V_BST); forced PWM, VMODE_H $\leq$ 1.2 V valid once VOUT $\geq$ 2.2 V (start-up pass-through holds VOUT near VIN).
C7 22 uF CIN at VIN/GND. C8-C11 4x 22 uF X6S 0603; -20-28 uF effective at 5.15 V, inside TI's 10=50 uF; TI 6.3 asks $\geq$ 10 uF effective up to 1.5 A and $\geq$ 20 uF from 1.5 to 3 A, met only typically, so the published BEC is 1.5 A continuous (D79: HD load 1.43=1.48 A, LED strip loaded in HD mode; V1 load step at 1.7 A). Feed-forward cap (TI 8.2.2.4: above 40 uF effective, fFZ 2 kHz = 1.7 nF across R4); C8-C11 give 20=28 uF, HD mode adds C12/C13 (-6 uF via the mux) and the O4 input (-47 uF, 20 uF): none fitted (lean). V1b steps the load in HD with a real O4, 1.5=2.2 nF across R4 only if it rings. L1 0.68 uH, Isat 6.5 A, 17 mOhm; valley limit 6.5=10 A; converter capability 1.67/1.82/1.98 A at 2.8/3.0/3.2 V (spec 4.2).
Passes VIN to VOUT when VIN $\geq$ VOUT (surge path, spec 4.1).
Layout (TI 10.1): Cout VOUT/GND pads at pins 3/1 (hot loop SW-VOUT-Cout-GND smallest). CIN at pins 7/1. SW copper minimal. R4/R5 at FB pin 4 away from SW; ground plane under the part.

04 LITE HD SUPPLY SWITCH TPS22810



New block from TI SLVSDH0C (D15, D33, QD5 B-10/22/24), U5 = TPS22810DRVR WSON-6 2x2 (DB3, P4 r4; smallest package, 3 A at TA 65 C vs 2 A DBV, RthJA 74.6 vs 182 C/W; same die and EN/UWLO figures: pins 1 VOUT=2 QD3 B-3 CT 4 GND/5 EN/UWLO 6 VIN, EP 7 TO GND as on TI's EVM, SLVUAY7B Fig. 1).
P4-2 / BR-10: U5 VIN on +5V_BST, ahead of the mux: USB never reaches the O4. No separate CIN (C17 deleted, P4 critique): the boost outputs C8-C11 (20-28 uF effective) are U5's input capacitor (TI 10.3: CIN optional, 1 uF usually sufficient); P5 condition: U5 VIN within 3 mm of C8-C11 or U3 VIN2 on the +5V_BST copper. C16 CT 10 nF: -4.7 V/m, 0.2 A inrush per 47 uF. Q4 tied to VOUT: the internal RPDMT empties +5V_HD when off.
U5_EN (bom_plan O4QED) holds exactly R9 33k / R10 18k from +BATT, R87 510k from +5V_HD, C129 100 nF, D2 anode and U5 EN/UWLO pin 5 (netlist checked), D2 BASI6LD cathode + R11 2.4k pull-down on HD_EN (GPIO27, PINO1).
P4-1 (VENR 1.13-1.30 V, VENF 1.08-1.18 V, +5V_HD 4.93-5.37 V): EN = 0.345 x cell (+0.0223 x +5V_HD when on). Nominal R: on at 3.27-3.77 V, dead at 2.78-3.10 V; 1 % R: on at 3.78-4.00 V, dead at 3.10-3.40 V; hysteresis 0.32 x 0.340 V, so the 0.31 x cell rebound + 1.4 V margin is OK.
GPIO27 low or undriven: EN <= VF 0.715 V + 66 uA 2.4k = 0.87 V < VENF 1.08 V, also with +5V_HD still up. GPIO27 high: D2 reverse >= 1.7 V; 1 uA hot leakage moves EN 11 mV (shed 0.03 V). R11 draws 1.4 mA.
Filter: 11.4 kOhm x 100 nF = 1.14 ms (140 Hz): the 96 kHz ESC ripple (0.21 V p-p at EN) falls to 0.3 mV, D2 + R11 discharge it below VENF in 0.25 ms; turn-on from the clamp -1.6 mV at 4.2 V, USB only (back-fed 2.5-2.8 V + 0.3 V ripple). EN <= 1.07 V < VENR, EN <= 1.62 V << 1.8 V abs max.

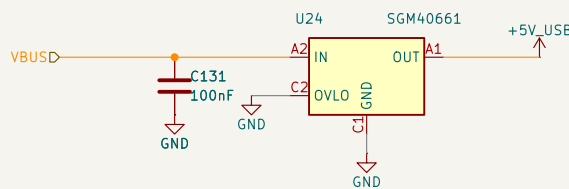
CAMERA 5 V FILTER



Source: OpenFC-Lite-Mini/hardware/power.kicad_sch (camera 5 V feed), with the commons 120 Ohm 0201 bead (blocks/LDO_IMU_TPS7A203S).

FB1 BLM03PX1225N1D: 120 Ohm at 100 MHz, 0.9 A, 160 mOhm: the 0.12 A camera drops 19 mV; it keeps the boost's 1 MHz and the camera's own switching apart. C15 10 uF X6S after the bead. +5V_CAM feeds the CAM pad J24 and the camera plug J32 (PAD5); PWR_FLAG because FB1 is passive. Feedback BLM03PX2205N1D (1.45 A).

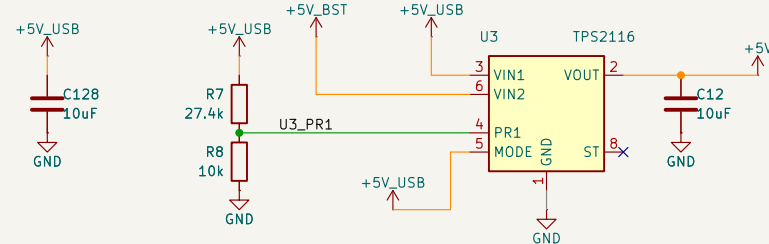
USB INLET OVP SGM40661



New block from the SGMICRO SGM40659/60/61 data sheet REV.A (D57).
 IN = VBUS (J31 pin 4, PADS). OUT = +5V_USB. OVLO (C2) to GND selects the internal threshold:
 5.75–6.12 V rising, 5.45–5.66 V falling, IN 28 V abs max, 2.7–20 V recommended (feature limit: 40 V surge immunity), OVULO 2.4 V, 15 ms debounce, then a 2 ms soft turn-on: a cable LC ring (simulated <= 10.8 V at IN) never reaches TPS2116 VIN1 (6 V abs max), and the soft turn-on is the VBUS surge limit USB asks for above 10 V downstream.
 C131 100 nF X65 25 V at the IN balls (data sheet CIN 0.1 uF; also takes ESD charge). COUT = C132 10 uF at U3 VIN1 (<= 100 uF allowed). RON 27 / 45 mOhm; 11–19 mV at 0.42 A. TA = 85 C.
 Open: a non-compliant source at 5.5–6.12 V would pass before the OVLO trips (TPS2116 VIN1 5.5 V recommended, 6 V abs max); USB sources are <= 5.25 V (5.5 V with PD).

Open: a non-compliant source at 5.5–6.12 V would pass before the OVLO trips (TPS2116 VIN1 5.5 V recommended, 6 V abs max); USB sources are ≤ 5.25 V (5.5 V with PD).

5 V POWER MUX TPS2116



New block from TI SLVSFG1A, priority mode (D24). C128 = the VIN1 cap: place it at U3 pin 3 (TI 9-10).
MODE tied to VIN1 (+5V_USB), PR1 = VIN1 x 10k / 37.4k; USB is selected above VREF x 3.74 = 3.44-4.04 V (VREF 0.92-1.08 V;
3.37-4.10 V with 1% x R), +5V_BST (VIN2) otherwise. ST (open drain) unused.
P4-2: +5V_BST holds only U2 VOUT/MODE, C8-C11, R4, U3 VIN2 and U5 VIN (netlist checked). USB selected: VIN2 stays off (truth
table 'VIN2 = X') and both channels block reverse current, so USB never feeds +5V_BST or the O4.
P4-3: C128 10 uF x65 0402 at VIN1 (TI CIN 1 uF); VIN1/MODE/PR1 peaks are bounded by U24. Unplug with a cell: +5V follows
VBUS to the PR1 threshold, then the 8 uS break-before-make gap: dV = 0.4 V x 8 uS / (C12 + C13 + C15, 3.4 uF each at 3.4
V, Murata curve) = 0.55 V -> +5V => 3.19 V typical; worst corner (3.44 V, -20% caps) 2.75 V. +3V3 stays >= 3.07 V. It
L12912 blocks RCB current and its >= 7 uF output carries 0.2 A for 0.8 uS. V1 case (b) scopes it.
C129 10 uF VOUT = +5V hold-up. MUX 40/60 mOhm: <= 0.04 V at 0.70 A analog; HD about 0.25 A (the O4 is on +5V_BST ahead
of the mux, D53 R8-R9).
TI 7.3.4 (clamp VOUT or about 100 uV where RCB occurs): RCB happens by design at USB plug/unplug with a cell; considered
deviation, no part (short +5V copper, near-zero current at the 42-70 mV trip). V1: VIN1/VOUT <= 6.0 V.

SHEET NOTES

POWER sheet, generated with hardware/tools/sch_build.py + sch_root.py (P4 power sheet), do not edit by hand.
R15 are power symbols (net = Value): +BATT_IN, +BATT, +5V_BST, +5V_USB, +5V, +5V_CAM, +5V_HD, +3V3, GND. Sheet pins: HD_EN (in), CURR_SENSE (out), VBUS (in). Local nets: SHUNT_SENSE_P/N(RC), U2_SW, U2_FB, U3_PR1, U5_CIN.
Closed here: P4-1, P4-2, P4-3, P4-7, P4-9, P4-12 (+3V3), P4-17 (BR-14 C13), P4-19 (C30 deleted); R11 is the GPIO27 pull-down P4-4 counts on; P4-13 is void (D9; C2, C5, C14 stay X5R, sized on DC bias). Values: research/DESIGN-SPEC.md, 4.1, 4.2, 15.5; P4 hand calculations reproduced in the shard notes.
P4 (ride note) (0.72V) C30 (ESC3 CFMB VDD bulk) deleted; ESC now has one 10 uF hot-loop cap per AGM210M cap on its own sheet, and ESC3's slot-M cap C35 sits within 4 mm of UB VDD pin 4 (BR-05 as amended, 100 nF C31 at the pin). C17 (U5 CIN) deleted; C8-C11 serve, P5 placement condition above.

OpenDrone

Sheet: /POWER/
File: power.kicad.

Title: Power: input, current sense, 5 V, 3.3 V, 04

Size: A3

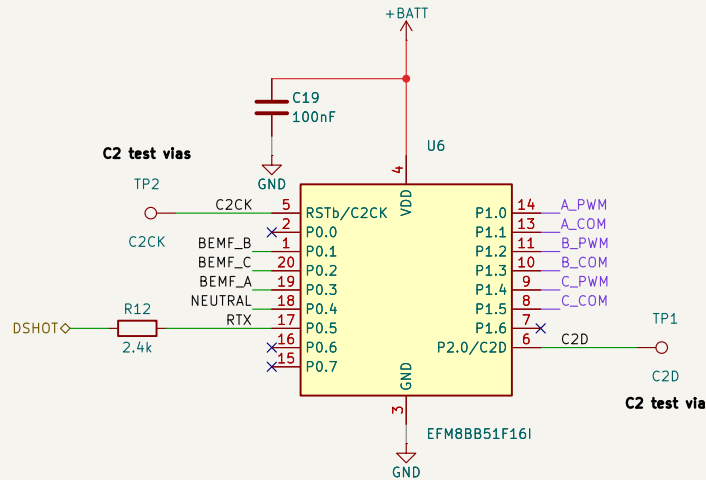
Date: 2026-10-09

Rev: rev1

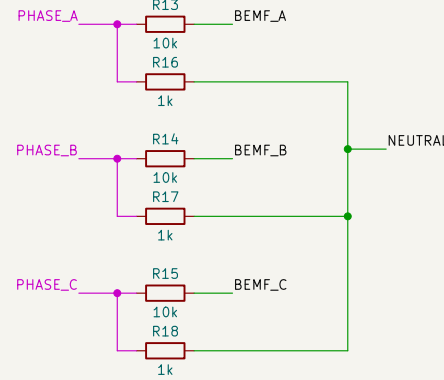
KiCad E.D.A. 10.0.6

id: 2/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



BEMF network: 10k series + 1k neutral star



ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12–R18, Q29–Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19–R25, Q32–Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26–R32, Q35–Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33–R39, Q38–Q40, TP7/TP8

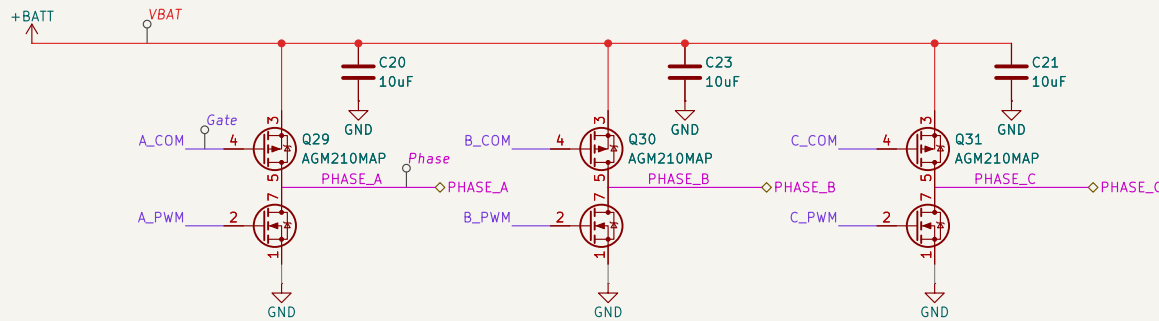
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail): setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4–11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered–FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered): V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle–high back–feeds +BATT through the EFM8 I/O diode, <= (3.3 V – VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



OpenDrone

Sheet: /ESC1/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4 Date: 2026–10–09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 3/14

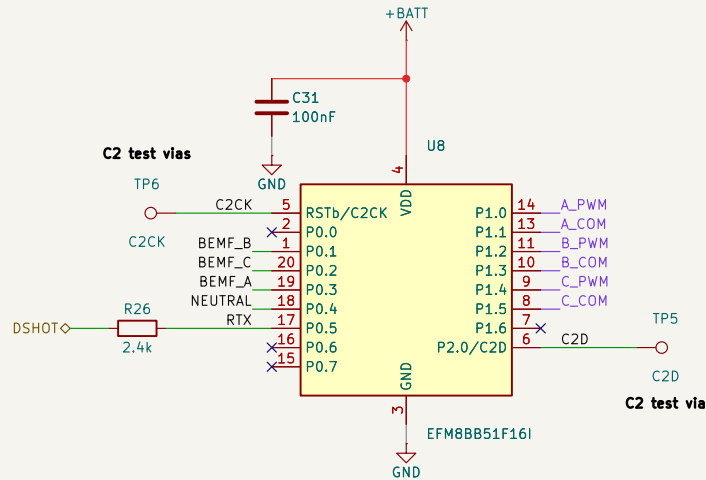
5



5

Id: 4/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



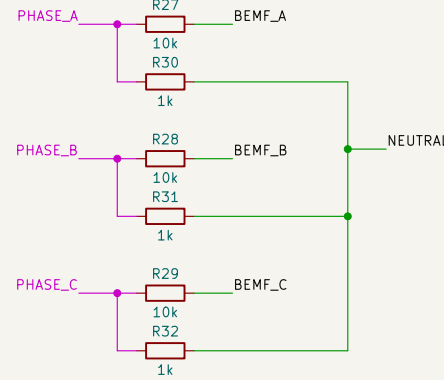
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX, P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D. Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources): C19 100 nF X6S at pin 4 / GND pin 3. The data sheet's 1 uF (Fig. 5.1) is the phase B hot-loop cap C23 10 uF X6S (about 2.6–2.8 uF effective at 4.2 V, Murata DC-bias curve); BR-05 as amended (P4 critique): within 4 mm of pin 4 through the +BATT plane, 100 nF at the pin (v3 cell site 3.4–3.9 mm). ESC3 has no extra cap (C30 deleted, D77). VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7–5.25 V (V3c). Ramp: DS Table 4.7 RAMPVDD >= 10 us/V vs about 1 us/V at a hot plug (spec 4.1): Matrix-parity deviation, V3c 20 hot plugs; a 2.2–4.7 Ohm VDD R (tau 0.5 us) does not fix it.

DSHOT (P4–B): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 – 2.5) V / 2.4k; about 0.1–0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP 12).

C2D / C2CK: TP1 / TP2, filled test vias (D67), bench recovery only: NextPCB pre-programs Bluejay; updates via Betaflight passthrough.

BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA, 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4–B: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

ESC1 M1 rear right: U6 C19, hot loop C20 C23 C21, R12–R18, Q29–Q31, TP1/TP2

ESC2 M2 front right: U7 C25, hot loop C26 C29 C27, R19–R25, Q32–Q34, TP3/TP4

ESC3 M3 rear left: U8 C31, hot loop C32 C35 C33, R26–R32, Q35–Q37, TP5/TP6

ESC4 M4 front left: U9 C37, hot loop C38 C41 C39, R33–R39, Q38–Q40, TP7/TP8

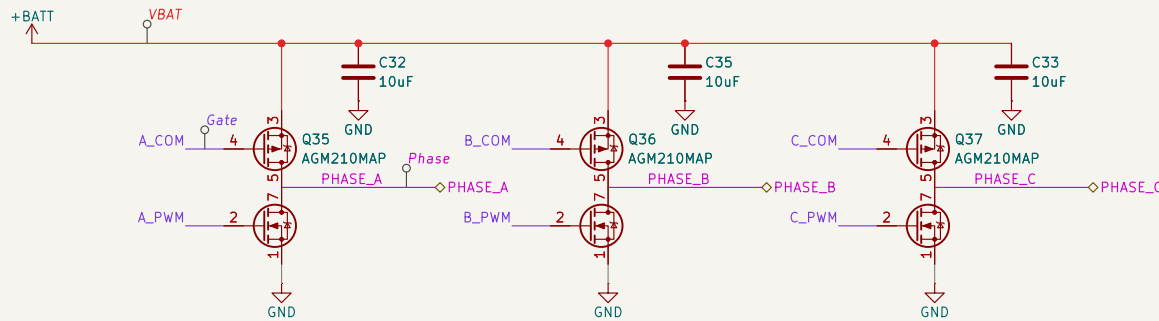
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail): setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4–11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70–220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1–2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA: 0.72 V over the 3.63 V unpowered–FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered): V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle–high back–feeds +BATT through the EFM8 I/O diode, <= (3.3 V – VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. Hot loop (D77, P4 critique): one cap per AGM210MAP, C20 / C23 / C21 10 uF X6S 0402 from pin 3 S2 (+BATT) to pin 1 S1 (GND) at the cell's lead side, both pads via-in-pad, so the commutation loop closes at the pins (Matrix II: one cap per FET); the three are also the ESC +BATT bulk, Bluejay: first power-up A_X_10_96 (about 200 ns; DS td(off) + tf = 58 ns N at RGEN 3.3 Ohm, 84 ns P at 1 Ohm, longer on EFM8 port drive), then A_X_5_96 (Matrix II) after the V2 shoot-through check (PINMAP 7.3); the DS Qg of 23 / 33 nC disagrees 3–5x with Ciss 950 pF, so V2 measures the gate edges.

OpenDrone

Sheet: /ESC3/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

Date: 2026–10–09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 5/14



C



Id. 6/14

R

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, $I_{POR} < 390$ μ A) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV fuel), through $\approx 72k$ (pull-up + R12), i.e. ≈ 60 μ A: 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix pathy (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the P10 DShot idle-high-back-feeds +BATT through the EFM8 I/O diode, $\leq (3.3 \text{ V} - V_F) / 2.4k$, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

OpenDrone

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

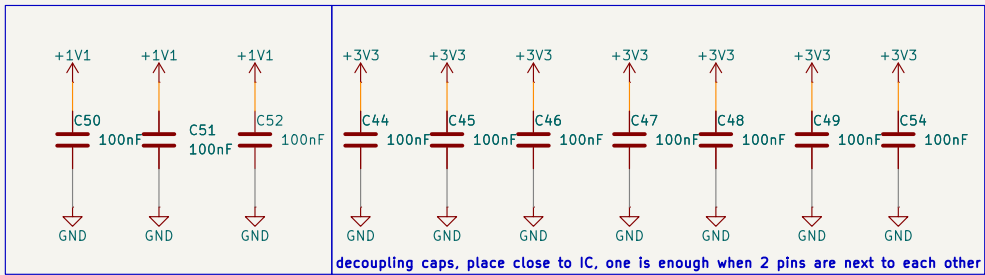
Size: A4	Date: 2026-10-09
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KiCad F D A 1006

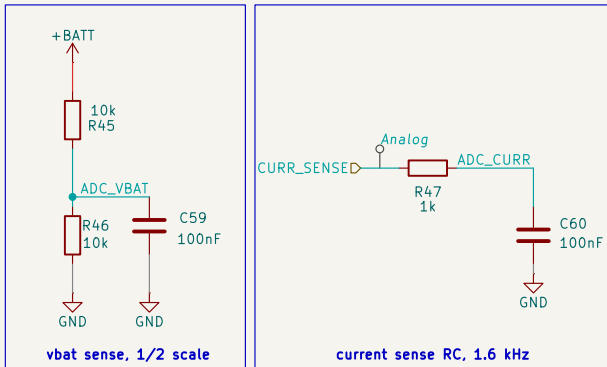
Rev: rev1

Id. 6/14

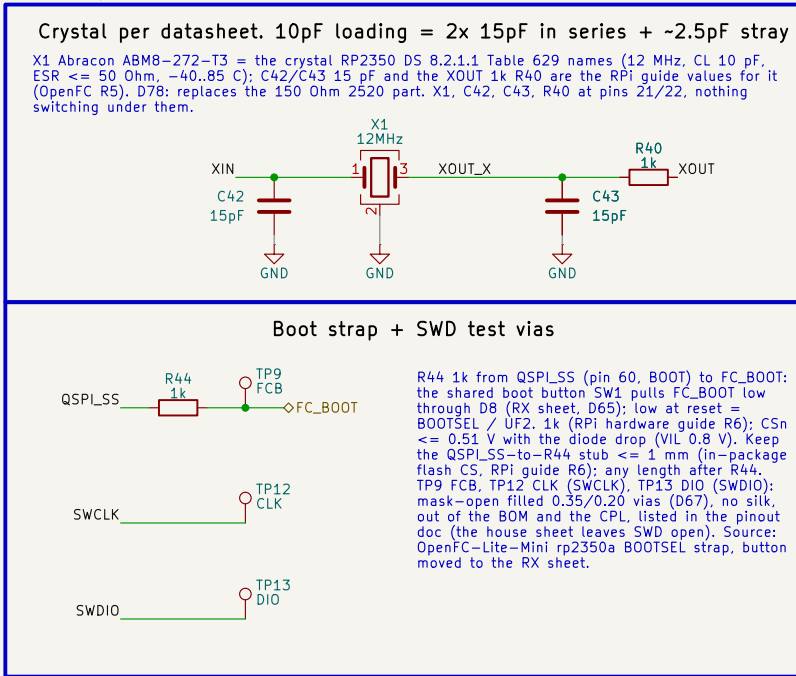
RP2354A: Smaller variant with integrated flash (2MB)



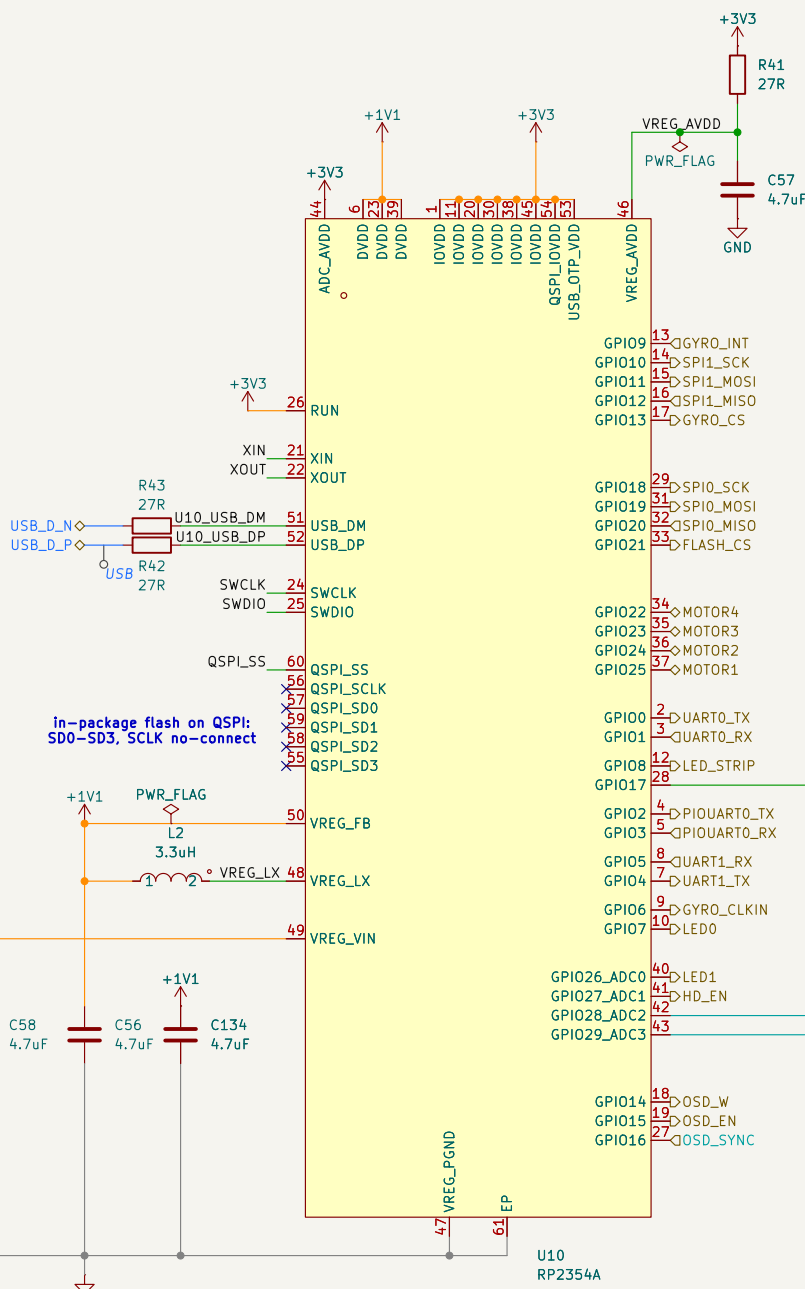
DVDD 6/23/39: C50–C52. IOVDD 1/11/20/30/38: C44–C48, one per pin. C49 sits at ADC_AVDD pin 44 and also serves IOVDD 45 (BR–08: RP2350 DS 6.1.5: 100 nF close to ADC_AVDD). C54 $\leq$ 1 mm from QSPL_IOVDD pin 54 (the in-package flash supply, 2.97–3.63 V) and also serves USB_OTP_VDD 53 (BR–08, P4–17 placement condition). All Murata GRM033C91E104KE14D 100 nF X6S 0201 (BR–18). Copied from OpenFC–Lite–Mini rp2350a (7 caps for the 9 3V3 pins there too); its 4.7 uF on this row moved to the SMP5 block (C56).



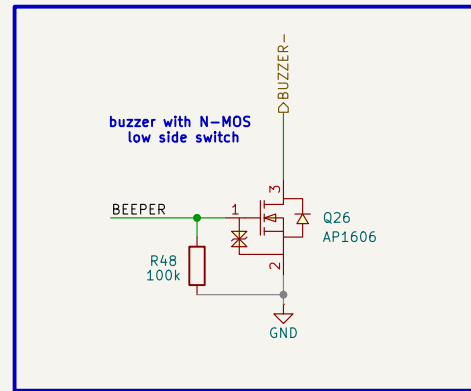
VBAT: R45/R46 10k/10k (house 100k/10k), 4.35 V $\Rightarrow$ 2.18 V on GPIO29 (ADC3, not 5 V tolerant). vbat_scale 200/100; 5 kOhm with C59 = 318 Hz. CURR: R47 1k + C60 100 nF = 1.6 kHz on GPIO28 (ADC2); INA186A3 on POWER; 50 mV/A. Ibatascale 500. Copied from OpenFC–Lite–Mini rp2350a (vbat + ESC current).



R42/R43 27R within 1–2 mm of pins 52/51 (RPI HW design 5.1, house 30R). USB_D_P/N: 90 ohm pair over unbroken GND, length-matched (D53); D7 ESD + J31 on PADS. Pin-side nets. U10_USB_DP/DM.



RP2350 1.1V DC–DC SMP5: follow the datasheet placement; RP2350 DS 6.3.8 (binding): C58 = CIN at VREG_VIN, C56 = COUT on +1V1 with VREG_FB taken from the C56 pad (not under L2); C58, L2 and C56 on the U10 side, smallest hot loops; CIN/COUT GND to VREG_PGND at one point with 2 vias; no copper under L2 / VREG_LX on the next layer. L2 AD1A–B201610S3R3 is polarity-marked; orientation as DS Fig. 26 / 28 (pin 2 = VREG_LX as on the flown house board). P5 condition (tools/check_p5_conditions.py SMP5): C58, L2, C56 and C57 on U10's side within 3 mm of pins 49 / 48 / 50 / 46. R41 27R + C57 4.7 uF filter VREG_AVDD (1.25 kHz, BR–21: 3.135–3.63 V); C57 gets its own GND via to the EP. D5: CIN $\geq$ 4.7 uF, COUT 4.7 uF +/-20 %; C134 = the 2nd 4.7 uF on +1V1 that DS 6.3.8 recommends at DVDD pin 23 (fitted per the 2026–10–09 ruling, house board fits two); place it at pin 23.



Q26 AP1606 (copied from OpenFC–Lite–Mini rp2350a, Q5 there); 20 V, 0.7 A, Vgs(th) 0.4–1.2 V, RDS(on) $\leq$ 220 mOhm at Vgs 2.5 V (ALLPOWER D5 p2). GPIO17 high = on (BEEPER_INVERTED, active buzzer, no PWM); R48 100k holds the gate low through reset. BZ+ = +5V and BZ– = drain on the GND LED 5V BZ– pad row (PADS).

SHEET RP2350A, source: OpenFC–Lite–Mini/hardware/rp2350a.kicad_sch (flown Rev2), copied block by block with its layout, frames and text style. Changes: USB–C + CC removed (USB = JST SH J31 + D7 on PADS; 27R series R42/R43 stay at U10, D52 BR–21 value); I2CO pull-ups removed (GPIO4/5 = UART1); the inverting WS2812 stage is not copied (F1, BR–04; GPIO8 drives the LED pad); status LEDs moved to the LED sheet; BOOTSEL button replaced by R44 + FC_BOOT (shared button on RX); SWD to test vias (D67); VBAT divider 10k/10k; 12 MHz crystal = ABM8–272–13 (D78); VREG_AVDD 27R; two 4.7 uF on +1V1 (C56 at VREG_FB, C134 at DVDD 23). GPIO map: research/PINMAP.md 2–3 (30 of 30 GPIOs, no move); Betaflight config PINMAP 5.

gyro SPI1 (IMU: GYRO_CS pull-up R49 there)

blackbox SPI0 (FLASH_CS pull-up R50 there)

PI00 bidir, DShot to the EFM8s (2.4k at the EFM8)

for CRSF receiver (ESP32 ELRS, UART0)

WS2812 pad LED, driven direct (F1, BR–04)

for external: user pads TP0/RP0 (PIO1, 8N1)

for MSP DisplayPort: DJI O4 Lite TX1/RX1

gyro CLKIN (ICM–42688–P only)

status LEDs (LED sheet, sinking)

HD line PINIO1: high = 04 on, analog VTX off

PIO2 framebuffer OSD: W, EN, SYNC consecutive

P4–11 power-up and 5 V-tolerant pads (PINMAP 4.5, F16: RP2350 DS Table 1433); an FT pad takes $\leq$ 3.63 V while IOVDD is 0–1.8 V (4.2 V at 2.5 V, 5.5 V at 3.3 V); the ADC pads GPIO26–29 take $\leq$ IOVDD + 0.5 V.
– MOTOR1–4 (GPIO25–22, F1), battery first: each EFM8 weak pull-up (70–220 kOhm to VDD = cell) reaches the pad through its 2.4k before +3V3 is up (boost soft start + LP5912 ramp, about 1–2 ms). Worst case 4.35 V open circuit = 0.72 V over 3.63 V, from $\geq$ 72 kOhm ($\leq$ 60 uA). Matrix parity (G473 FT pins –4 V at VDD 0). V9 scopes it: fallback (F16): 3.3 V-domain clamp or DShot-path enable.
– ADC_VBAT (GPIO29): R45/R46 put up to 2.18 V on the pad from 5 kOhm while IOVDD $<$ 1.7 V, so the pad clamp to IOVDD conducts $\leq$ 0.3 mA during the ramp; House + Matrix parity; V9 scope.
– HD_EN (GPIO27): R11 2.4k + the BA516LD from O4_EN ($\leq$ 90 uA) hold it at $\leq$ 0.22 V before and during reset; inside IOVDD + 0.5 V and below the AP1606 HD-gate Vgs(th) min 0.4 V at 25 C (P4–4: about 0.25 V at 85 C, –2.5 mV/K; a hot low-vth part may hold the analog VTX off until PINIO1 is driven low, R68 47k, V9 hot-soak boot check); driven high it is $\geq$ 2.62 V (VOH, 1.4 mA into R11) over Vgs(th) max 1.2 V.
– UART0, SPI0/1, gyro, OSD, CURR_SENSE: every driver is on +3V3 = IOVDD, no excursion. UART1 TX1/RX1: the O4 is fed only through U5 (needs HD_EN high), none. USB D+/D–: the host drives only after the device pull-up.
– PIOUART0 pads TP0/RP0, and TX1/RX1 used as the second hardware UART in analog builds (IO–SEI row 8); a peripheral on the 5V pads is up before IOVDD; 3.3 V logic only (README); a 5 V TX on RP0 or RX1 (GPIO5) would exceed 3.63 V at power-up.

OpenDrone

Sheet: /RP2350A/
File: rp2350a.kicad_sch

Title: RP2354A flight controller MCU

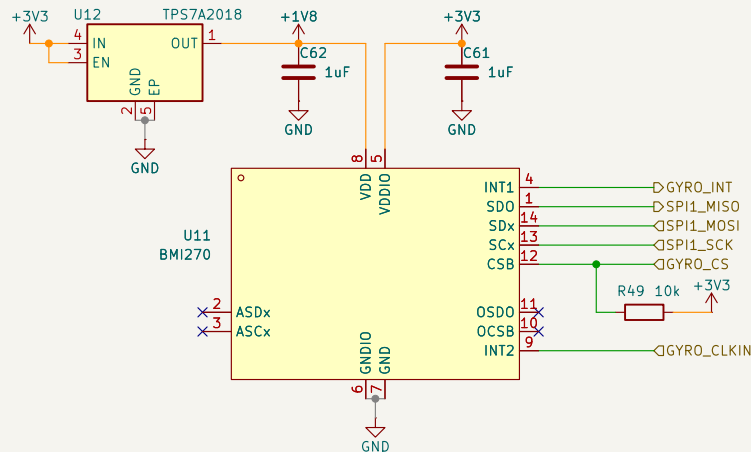
Size: A2 Date: 2026–10–09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 7/14

Universal IMU wiring and LGA-14 land (BMI270, ICM-42688-P)



GYRO BMI270: U11, R49

Source: OpenFC-Lite-Mini imu.kicad_sch (universal LGA-14 land, SPI1 4-wire, R49), copied and adapted. BMI270 on the BOM; TDK ICM-42688-P qualified on the same land (V10 fly-off, one part per revision).

Pins 2/3 (BMI270 ASDx/ASCx, TDK RESV) and 10/11 (OCSB/OSDO, TDK RESV): no-connect. BMI270 DS Table 22: unused ASDx/ASCx go to VDDIO or stay open, never to GND (the house sheet grounded them for the LSM6DSV16X).

Pins 6/7 GND (TDK pin 7 RESV must be GND). VDDIO pin 5 = +3V3 (RP2354A I/O level). VDD pin 8 = +1V8; the BMI270 has no supply-order rule (DS 4.3). Abs max 4.0 V on VDD/VDDIO, I/O up to VDDIO + 0.3 V.

Pin 9: TDK CLKIN = GYRO_CLKIN (RP2354A GPIO6, PWM 32 kHz, driven only with the TDK part); BMI270 INT2, unused, left disabled. R49 10k; GPIO13 resets with its pull-down on, which would select the chip; R49 holds CSB >= 2.58 V (0.78 x VDDIO with the RP2354A RPD min 36 kOhm at IOVDD 3.3 V, RP2350 DS Table 1683) > VIH 0.7 x VDDIO = 2.31 V.

P5: top side next to U12, LGA rotated 90 deg: GYRO_1_ALIGN from the placed orientation (PINMAP F11).

+1V8 GYRO LDO: U12, C61, C62 (D52 BR-09, P4-17)

Source: pcb-agent-commons blocks/LDO_IMU_TPS7A2033 (OpenFC-H7) topology with the 1.8 V part and no ferrite (fed from the +3V3 LDO): the house OpenFC-Lite-Mini also runs the gyro VDD from its own 1.8 V LDO. TPS7A2018 DQN: 1 OUT, 2 GND, 3 EN, 4 IN, 5 EP to GND. IN = EN = +3V3; VIN 1.6-6.0 V, VEN 0-6.0 V (abs max 6.5 V), EN never above IN. Load < 1 mA.

No 100 nF at the gyro: C61 = U12 CIN at BMI270 VDDIO pin 5, C62 = U12 COUT at VDD pin 8 (COUT >= 0.47 uF effective, ESR <= 100 mOhm; about 0.6-0.85 uF at 1.8 V). C61 counts in the +3V3 COUT sum (P4-12, POWER).

P5: U12 on the gyro side with OUT facing pin 8; C61 between U12 pin 4 and gyro pin 5, C62 <= 1.5 mm from pin 8 (P5-06), else re-place U12.

OpenDrone

Sheet: /IMU/

File: imu.kicad_sch

Title: Gyro BMI270 + 1.8 V LDO

Size: A5

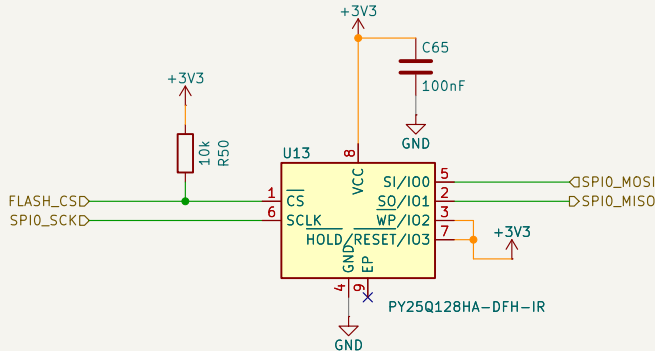
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 8/14

SPI NOR PY25Q128HA: 16 MB blackbox on SPI0 (U13 C65 R50)



Placement (P5): beside the RP2354A on SPI0 (bom_plan region nor, top; either side works). U13, SPI0 and FLASH_CS stay ≥ 3 mm from the RX antenna hole (spec 4.9 and 11, DRU 'SPI0 3 mm from the RX antenna hole'): SPI0 harmonics fall in the ELRS band, V7 measures desense while logging. C65 at pin 8. Filled+capped 0.35/0.20 via-in-pad allowed in the lead lands, never in the tabs. Betaflight sets SCK to fast slew (bus_spi_pico.c); MOSI and CS keep the reset 4 mA drive and slow slew (EMI lever without parts). Sourcing: no LCSC listing; NextPCB buys by MPN (D60), the quote confirms it. Same-pinout fallbacks in the Betaflight table (USON-8 4x4 land not staged): PY25Q128HA-QVH-IR, GD25Q128EQIGR.

Source: OpenFC-Lite-Mini hardware/blackbox.kicad_sch (rev2). Kept: layout, the 10k CS pull-up (R39 $\rightarrow$ R50) and the 100n supply cap (C43 $\rightarrow$ C65). Changed: its microSD (Card1) became this 16 MB SPI NOR (D66); the SPI0(SCK,MOSI,MISO) bus label became four hierarchical labels (sch_contract.json).

U13 Puya PY25Q128HA-DFH-IR: 128 Mbit, USON-8 3x3 (DF), 2.7-3.6 V, -40..85 C, fC 133 MHz, READ 03h 80 MHz, JEDEC ID 85 20 18 is in the Betaflight m25p16 table: USE_FLASH_M25P16, SPI0 PA18/PA19/PA20, FLASH_CS_PIN PA21 (PINMAP). SPI0 runs at 75 MHz for writes AND reads: Betaflight bus_spi_pico.c spiCalculateDivider() uses an even prescale ≥ 2 on clk_peri 150 MHz, so the table's 133 / 80 MHz both give 150/2. Writes are inside fC; reads are not guaranteed: tCLQV 6-7 ns vs the 6.67 ns SCK half period plus pad and trace delay (the RP2350 DS gives no SPI receive timing). Gate F7 / V9: erase, write, read back and CRC a full log, status polling under load; fallback = a lower read clock for this JEDEC ID (firmware cap or table entry), upstreamed with the board config.

WP# (3) and HOLD#/RESET# (7) tied to +3V3: the DS gives an internal pull-up only to the SOP-16 dedicated RESET#, so pin 7 must not float. Delivered SR = 00h, CR = 00h (QE 0, HOLD# function, WPS 0: no block lock) and Betaflight sends 1-bit SPI commands only, so IO2/IO3 never drive against the rail (the ERC pin_to_pin warning on pins 3/7 against a +3V3 power output is this deliberate tie).

R50 10k: CS# must follow VCC at power-up (DS 5.5). RP2354A pads reset with the pull-down on (PDE 1, RPD 36-113 kOhm at 3.3 V): CS $\geq 36/46$ VCC = 0.78 VCC > VIH 0.7 VCC, so U13 stays deselected through reset and boot. U13 and the RP2354A IOVDD share +3V3: no pin above VCC + 0.5 V.

C65 100 nF X65 at pin 8, GND return straight to pin 4 (ICC max 13 mA read, 12 mA erase; adds 0.1 uF nominal to the +3V3 COUT sum of P4-12). EP (9) = two tie-bar tabs with no electrical function in the DS: no-connect, no vias in the tabs.

OpenDrone

Sheet: /BLACKBOX/

File: blackbox.kicad_sch

Title: Blackbox 16 MB SPI NOR PY25Q128HA

Size: A5

Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 9/14

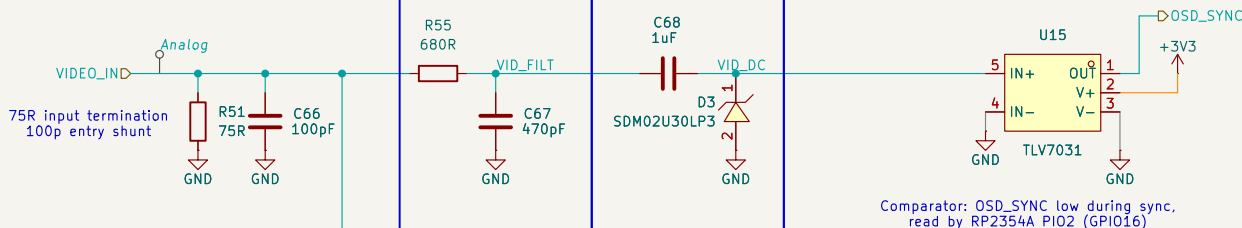
VIDEO PATH

Chroma Filter
fc = 500 kHz

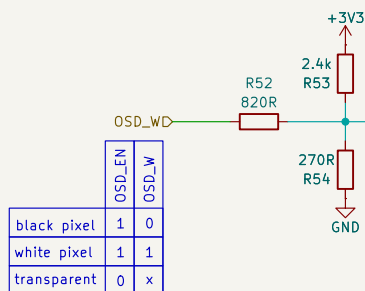
DC Restoration

Sync detection, VID_DC sync tips
clamped to $-V_F$ (about $-0.1...-0.2$ V)

U15 V+ (2) is decoupled by C70 in the switch cell,
one 100n shared with U14 (BR-13)



Resistor divider lets the OSD_W GPIO select
white (high: 1.05 V) or black (low: 0.27 V)



S low (OSD_EN low / reset): B1 camera
S high: B2 OSD level

Analog switch

VIDEO_OUT: the switch output drives the RTC6705 video network on the VTX sheet directly (R58 1.2k || C104 33p, R59 270R, C105 4.7u, R60 1.2k: OpenOSD-X values); the house COS8051 buffer (U1, R46-R48, C41) is deleted (spec 4.8). OSD levels (P4-10, recomputed at the P4 critique with the real load): open-circuit at OSD_LVL (+3V3 3.30 V, OSD_W push-pull) black 0.268 V / white 1.054 V from 195R (R53) | R54 = 0.352 V / 256R, with R52) + switch on 7-18R. VIDEO_OUT is loaded by the VTX network ahead of C105: R58 + R59 = 1.47 kOhm at DC, about 0.95 kOhm at 5 MHz and 0.60 kOhm at 10 MHz (R58 / C104 4.0 MHz, 21.9 MHz with R59; 270R only above about 20 MHz). Loaded levels: black 0.234-0.236 V, white 0.920-0.927 V (swing x0.88 of the open-circuit 0.786 V); edge gain x0.86 at 5 MHz and x0.80 at 10 MHz vs x0.97 for the camera (37.5R source): OSD text slightly dimmer and softer than the camera picture, not a failure (the house COS8051 buffer that made the source low-impedance is deleted). V9 checks OSD white / black and edge sharpness on a monitor; fallback: rescale R52-R54 within the 3 mA OSD_W budget. PS-01 (R54 270R) would lower white to about 0.89 V: not recommended. OSD_W: 2.74 mA high (≤ 3 mA, 2.79 mA at 3.36 V), 0.33 mA low.

Source: OpenFC-Lite-Mini hardware/osd.kicad_sch (rev2, flown), copied and adapted: cells, texts, truth table and part spots kept (A3 -> A4). Same topology: 75R termination, 680R/470p chroma filter, 1u + SDM02U30LP3 sync-tip clamp, TLV7031 comparator, SN74LVC1G3157 camera/level switch (contract source, spec 4.8).

Changed: COS8051 buffer deleted (output = VIDEO_OUT); divider R42 3.9k / R43 12k / R44 1.4k -> R52 820R / R53 2.4k / R54 287R (about 5x down, P4-10); C39 + C42 -> one shared C70 100n X65 (D52 BR-13 + BR-18); C66 100p entry shunt added (spec 11); refs per bom_plan (R41>R51, C38>C67, C40>C68, D9>D3, U2>U15, R45>R55, R42-44>R52-54, U18>U14). Symbols from the OpenDrone catalogue with TI pin numbers: TLV7031 X2SON 1 OUT, 2 V+, 3 V-, (centre pad), 4 IN-, 5 IN+, SN74LVC1G3157 DTB 1 B2, 2 GND, 3 B1, 4 A, 5 VCC, 6 S. The house lib symbols used the EasyEDA pad order (consistent with its own footprints), so the drawing is re-routed with the same nets: IN+ = VID_DC, IN- = GND, B1 = camera, B2 = OSD level. The house sheet's rotated ground is upright here.

Operation: OSD_EN low (RP2354A reset pull-down, OSD idle) -> B1, the camera passes through. OSD_EN high -> B2: OSD_W high = white, low = black. No OSD without camera sync (osd_pico.c overlays on detected sync). OSD_W/EN/SYNC = GPIO14/15/16, consecutive, PIO2 (PINMAP).

Sync threshold (P4-10): 680R/470p = 498 kHz: colour burst x0.11-0.14 (+-17-21 mV), 4.7 us sync kept (tau 0.32 us). C68 + D3 (cathode on VID_DC) clamp the sync tips to $-V_F$; the leakage-limited clamp current (uA average, tens of uA at the tip) gives V_F about 0.1-0.2 V over temperature (I: DS 0.37 V max at 10 mA / 25 C, 0.20 V at 10 mA / 125 C). IN- at GND puts the trip point V_F above the tip, 33-67 % of the 0.3 V sync, ≥ 0.08 V from tip and blanking after VIO +8 mV and VHYS 2-17 mV. OSD_SYNC is low during sync (osd_tx.pio waits for a low sync pin). IN+ at $-V_F$ stays inside abs max ($V_{EE} - 0.3$ V), at most 0.1 V under the ROC input floor ($V_{EE} - 0.1$ V), no phase reversal (DS 6.4.1). tpd 3 us on both edges (equal shift, as on the house board).

Ratings: +3V3 only (U14 1.65-5.5 V, U15 1.6-6.5 V). S: VIH 0.7 VCC = 2.31 V, edges ≤ 10 ns/V (fast slew pads). AC-coupled camera sync tips (-0.3 to -0.4 V) are below the switch ROC (0.VCC) but inside abs max (-0.5 V, 50 mA; 37.5R source), as on the house board: V9 checks. Supply: U15 0.9 uA, U14 ≤ 10 uA, divider 0.94-1.26 mA + OSD_W 2.74 mA. C68 X5R carries ≤ 0.6 V DC: no DC-bias loss (P4-13 hot-zone rule void, D69).

Placement (P5; P4-17 sheet note): bottom, region osd. BR-13: C70 is the only supply cap of U14 (VCC 5) and U15 (V+ 2): place it between the two parts (2.5 mm apart). GND via-In-pad; the P5 check lists its distance to both pins. U15 centre pad (3, V-) to GND through a via (TI 7.4). R51 and C66 at the video entry on the CAM pad J23 / plug J32 line: camera return into L2 at the CAM GND pad: video on L1/L6 over a plane with GND both sides, never L3 (spec 11). R55, C67, C68, D3 next to U15 IN+. Video parts away from the boost and its inductor (preference, D69; v2 floorplan 3.3-8.6 mm: V5 video test). Netclass Analog: VIDEO_IN (flag here), VIDEO_OUT, VID_FILT, VID_DC, OSD_LVL, OSD_SYNC by the setup_board.py patterns: OSD_W, OSD_EN Default. Sourcing: exact MPNs from bom_plan: LCSC empty where bom_plan has none (C66, R52, R54; open item A9-LCSC). PS-01 (R54 -> 270R RC0201FR-07270RL, shared with R59, -1 line): black 0.258 V, white 1.011 V, 187R, OSD_W 2.79 mA, also passes P4-10; proposed in contract-issues, drawn = bom_plan.

OpenDrone

Sheet: /OSD/
File: osd.kicad_sch

Title: PIO analog OSD front end

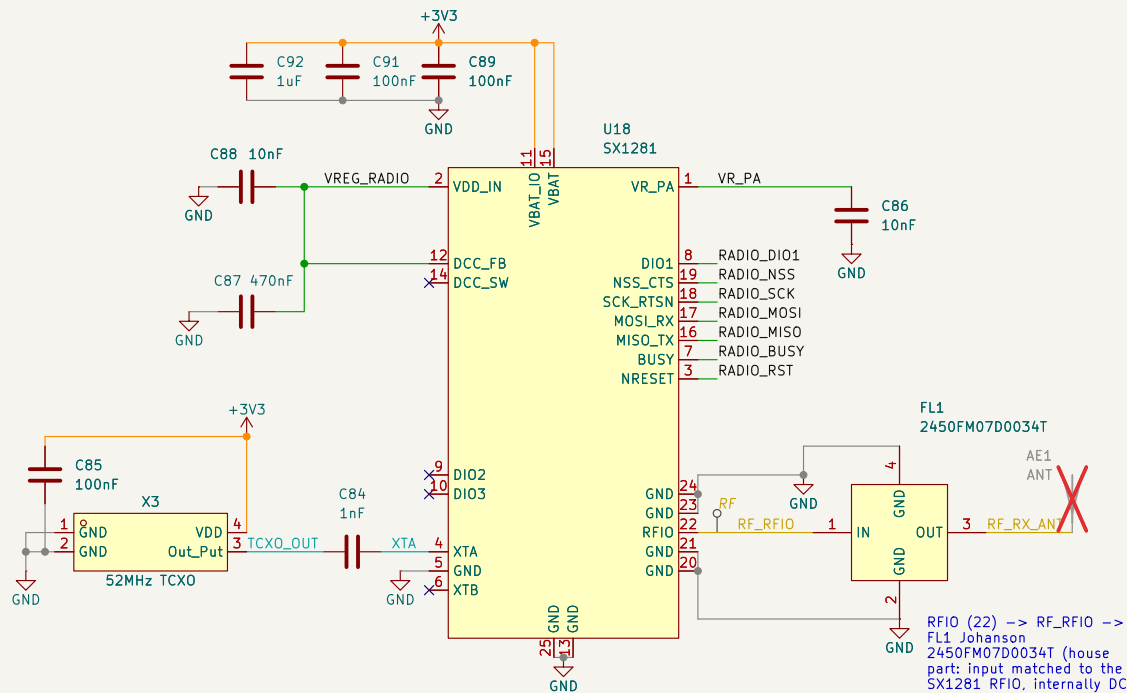
Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 10/14

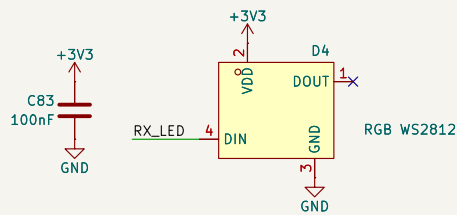
SX1281 RADIO + 2.4 GHz FILTER + WIRE ANTENNA



SX1281 (U18), copied from OpenRX-Lite-UFL; SX1281IMLRT (D52 BR-25), SX1280IMLRT drop-in alternate (ranging engine only).
LDO MODE (Semtech DS Rev 3.2 5.1, Fig. 15-4): DCC_SW (14) open, the sibling's 15 uH L4 deleted (the ELRS layout has no radio_dcdc); VDD_IN (2) tied to DCC_FB (12) = VREG_RADIO with C87 470 nF + C88 10 nF; VR_PA (1) C86 10 nF; VBAT_IO (11) C91 and VBAT (15) C89 100 nF (Table 15-1) + C92 1 uF radio local (P4-12 lever PS-06); the house sheet's extra 10 nF / 10 uF deleted (BR-06).
PIN 5 IS GND (DS pin table) and is grounded here; the sibling sheets left it open (only XTB stays open with a TCXO, DS 15.2). DIO2 / DIO3 open.
52 MHz TCXO X3 (kept, D53): clipped sine AC-coupled by C84 1 nF into XTA (4). C85 100 nF at VCC, XTB (6) open.

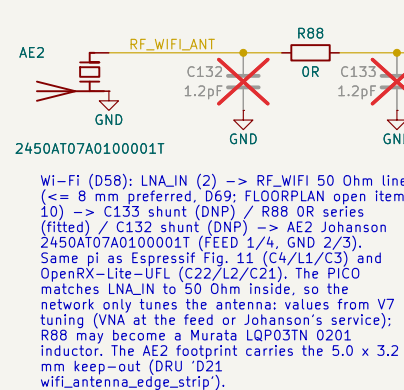
RFIO (22) -> RF_RFIO -> FL1 Johanson 2450FM07D0034T (house part: input matched to the SX1281 RFIO, internally DC blocked, IN pin 1 toward the chip) -> RF_RX_ANT (50 Ohm) -> AE1: insulated lambda/4 wire (about 28-29 mm, trimmed in V7) in a plated D0.5 hole; AE1 DNP, out of BOM / CPL. RF class: board rule areas RF_RX_ANT / RF_RX_EXIT.

ELRS RGB LED



D4 XINGLIGHT XL-1010RGB-2812B (copied from OpenRX-Lite-UFL) on +3V3, ELRS led_rgb GPIO22, one LED (overlay ledidx_rgb_vtx [0]).
P4-15 DECIDED: KEEP ON +3V3. Datasheet: VDD 3.5-5.5 V, DIN VIH >= 2.8 V, 5 mA per colour with a 0.6 V sink knee. On +3V3 the supply is 0.2 V under the table minimum (blue / green may run below 5 mA; Vf + knee is about 3.4-3.6 V); DIN comes from a 3.3 V CMOS output on the same rail. On +5V the supply would be in range but the data guarantee is short (ESP32 VOH >= 0.8 VDD = 2.64 V at a high-impedance load, DS Table 18, vs VIH 2.8 V). Neither option sits fully inside the tables; the house RX circuit (OpenRX-Lite, -UFL, OpenAIO) runs this LED on 3.3 V, so it stays. Bench V-LED on the first boards: red, green, blue and white at 3.3 V, cold and hot. Failback: VDD pin to +5V (one trace: C83 is 25 V).

Wi-Fi CHIP ANTENNA (D58)

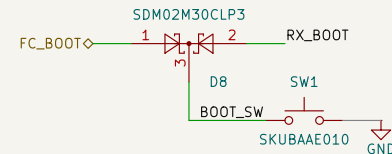


Wi-Fi (D58): LNA_IN (2) -> RF_WIFI 50 Ohm line (<= 8 mm preferred, D69: FLOORPLAN open item 10) -> C133 shunt (DNP) / R88 OR series (fitted) / C132 shunt (DNP) -> AE2 Johanson 2450AT07A0100001T (FEED 1/4, GND 2/3). Same pi as Expressif Fig. 11 (C4/L1/C3) and OpenRX-Lite-UFL (C22/L2/C21). The PICO matches LNA_IN to 50 Ohm inside, so the network only tunes the antenna: values from V7 tuning (VNA at the feed or Johanson's service); R88 may become a Murata LQP03TN 0201 inductor. The AE2 footprint carries the 5.0 x 3.2 mm keep-out (DRU 'D21 wifi_antenna_edge_strip').

NOTES

RX SHEET (P4 shard C). Sources: the SX1281 radio, 52 MHz TCXO, RGB LED and Wi-Fi pi match are COPIED from the house sheet OpenRX-Lite-UFL hardware/esp32c3_sx1281_lite.kicad_sch (same as OpenAIO rx_esp32c3_sx1281.kicad_sch); same symbol positions and wires, adapted to bom_plan refs/values/MPNs and to D52/D53/D58 (LDO mode, pin 5 to GND, U:FL replaced by the wire-antenna hole, 2450AT07A0100001T chip antenna, grounds turned down); the house Johanson 2450FM07D0034T front-end filter stays. The ESP32-PICO-V3 block is new, from the Expressif ESP32-PICO series datasheet v1.3 Fig. 11 (D52 BR-01, D55); the shared boot / bind button is new (D65, Matrix II style). Pin plan: research/PINMAP.md 6. Parts: hardware/bom_plan.json. Radio and LED +3V3 come from the POWER sheet (U4 LP5912-3.3).

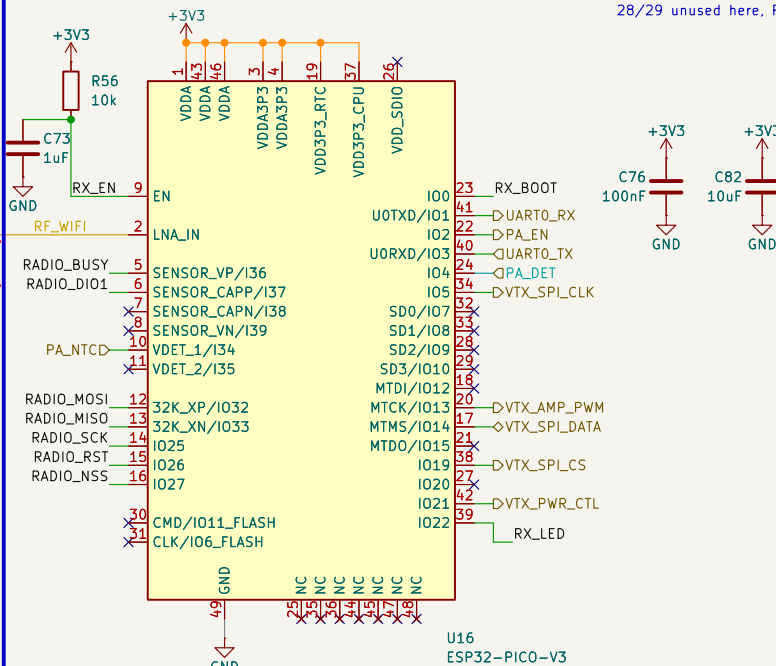
BOOT / BIND BUTTON (D65)



BOOT / BIND BUTTON (D65: new). SW1 Alps SKUBAAE010 (2.4 x 1.4 mm) from BOOT_SW to GND. D8 Diodes SDM02M30CLP3 dual Schottky, common cathode on BOOT_SW; A1 = FC_BOOT (RP2354A QSPILSS through R44 1k, RP2350A sheet), A2 = RX_BOOT (ESP32 GPIO0, pin 23).
Released: each strap sees only diode leakage (IR <= 0.15 uA at 10 V, 25 C), so GPIO0 keeps its internal pull-up and QSPILSS its own level.
Held while power is applied: RP2354A BOOTSEL and ESP32 ROM download mode (CSn <= 0.51 V vs VIL 0.8 V; GPIO0 <= 0.42 V vs VIL 0.825 V); each mode stays latched until that chip's next reset.
Production / ELRS recovery (D67, spec 12.1): (1) hold SW1 (or probe BOOT_SW) while powering; (2) write the Betaflight UF2 (or picotool reboot): the RP2354A restarts into Betaflight, the ESP32 stays in download mode; (3) esptool through Betaflight serial passthrough.
Pressed while running: ELRS button (bind 1.5 s, Wi-Fi 5 s, reboot 12 s); 2.9 mA through R44, CS stays >= 2.62 V, the FC keeps running.
Bench V-btn: if GPIO0 dips below 2.5 V while released (QSPILSS edges under XIP load coupling through D8), add 100 pF 0201 on BOOT_SW.

ESP32-PICO-V3 MCU (ELRS)

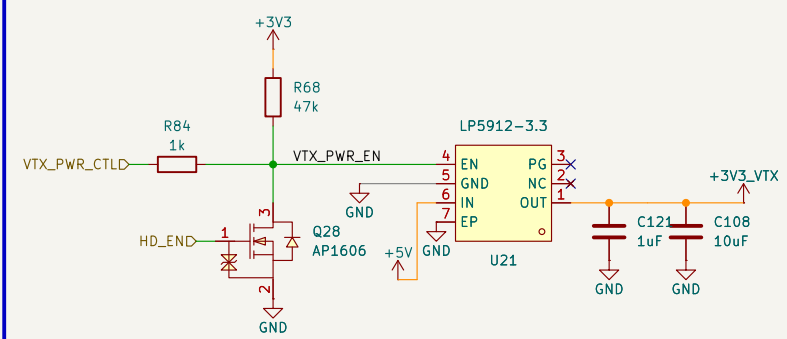
ESP32-PICO-V3 (U16), Expressif PICO datasheet v1.3 Fig. 11 / Table 4 (new block).
SUPPLY: VDDA (1, 43, 46), VDDA3P3 (3, 4), VDD3P3_RTC (19), VDD3P3_CPU (37) on +3V3; C82 10 uF + C76 100 nF at VDDA3P3 pins 3/4 (Fig. 11 C1/C2). The 4 MB flash, 40 MHz crystal, CAP1/CAP2 network and the VDD_SDIO capacitor are inside the SIP (Fig. 8): VDD_SDIO (26) stays open.
EN: CHIP_PU (9) = RX_EN, R56 10k to +3V3 + C75 1 uF (tau 10 ms), never floating.
STRAPS at reset (P4-6, checked on the netlist):
GPIO0 (23) RX_BOOT: internal 45k pull-up, D8 blocks -> 1 = SPI boot (SW1 held -> download).
GPIO2 (22) PA_EN: internal 45k pull-down + LP5907 EN 1 MOhm pull-down (VTX sheet), no external part -> 0.
GPIO5 (34) VTX_SPI_CLK: internal pull-up, R81 1k into the RTC6705 CMOS input -> 1 with +3V3_VTX up; reads 0 in HD mode (RTC6705 unpowered, its input clamps the pull-up to about 0.6 V), which only changes the SDIO-slave timing (unused).
GPIO12 / MTDI (18): no connect, internal pull-down -> VDD_SDIO 3.3 V for the in-package flash (VDD_SDIO eFuse burn optional).
GPIO15 / MTDO (21): no connect, internal pull-up -> boot log on U0TXD (FC ignores it).
NOT CONNECTED: flash CMD/IO11 (30) and CLK/IO6 (31); NC pins 25, 35, 36, 44, 45, 47, 48; free IO7/8/9/10 (32, 33, 28, 29; PSRAM on the -02 alternate), IO20 (27), I35 (11), I38 (7), I39 (8).
ELRS LAYOUT JSON (D53, D81): serialLrx 3, serialTx 1, radio_sck 25, radio_mosi 32, radio_miso 33, radio_nss 27, radio_rst 26, radio_busy 36, radio_dio1 37, led_rgb 22, button 0, vtx_nss 19, vtx_sck 5, vtx_mosi 14, vtx_miso -1, vtx_amp_pwm 13, vtx_amp_vpd 4, vtx_amp_vref 2. "vtx_miso": -1 is written explicitly: an overlay only replaces keys, so an omitted key keeps the stock 23 = the PICO flash DI, which SPIClass(HSPI).begin() turns into an input (flash cut, crash at VTX init); -1 makes the core attach the HSPI default GPIO12 (no connect here) instead, patched ELRS: GPIO12 VTX_PWR_CTL (+3V3_VTX LDO EN through R84 1k), GPIO34 PA_NTC (ADC1, read in the isTick window, erratum 3.11).
VTX LINES: series 1k R80/R81/R82 are drawn on the VTX sheet and placed at the ESP32 end (back-feed limit with +3V3_VTX off, F21). GPIO13 VTX_AMP_PWM = ELRS vtx_amp_pwm (RTC6705 PAOUT1 feed PNP, not the SE5004L bias). PA_DET is ADC2_CH0 (not readable with Wi-Fi on; not a flight state).
+3V3 LOAD: ELRS about 0.07 A in flight, Wi-Fi TX peaks 370 mA (DS Table 19, 802.11b 19.5 dBm) in ELRS Wi-Fi mode only (disarmed, bench); +3V3 peak about 0.48 A with the FC and LED, inside the LP5912 500 mA rating; LDO heat and the COUT sum belong to the POWER sheet (P4-12).
PLACEMENT: SiP site from floorplan v3 (D55, D61); no 85 C halo rule (D69). Alternate part ESP32-PICO-V3-02 (pins 28/29 unused here, P4-14).



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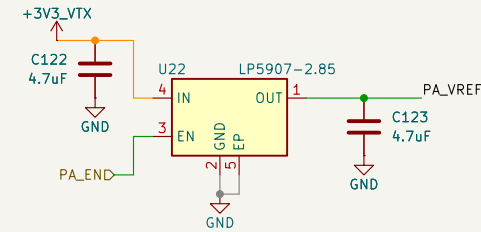
Sheet: /RX/ File: rx_esp32_sx1281.kicad_sch		
Title: ELRS 2.4 GHz RX: ESP32-PICO-V3 + SX1281		
Size: A3	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6	Id: 11/14	

+3V3_VTX LDO + HD GATE



U21 LP5912-3.3 on +5V; CIN = the shared 10 uF C13 on POWER (BR-14) only if P5 puts C13 within 1 cm of the U21 IN pin (LP5912 DS 9.2.2.2; tools/check_p5_conditions.py), else U21 gets its own 10 uF 0402 CIN (GHW15SC00106ME110). EN = VTX_PWR_EN; R68 47k to +3V3 (stock ELRS runs the VTX); R84 1k from ESP32 GPIO21 (patched-ELRS power-down hook); GPIO21 low gives VEN 0.07 V < VEN(OFF) 0.3 V; Q28 AP1606 drain = HD gate (HD_EN high: <= 220 mOhm at VGS 2.5 V sinks 70 uA, 3.3 mA with GPIO21 high; VEN < 1 mV). Hot margin (P4 r4, R68 was 100k); EN stays >= VEN(ON) 1.3 V max up to 42 uA of Q28 leakage (100k: 20 uA); AP1606 IDSS <= 1 uA at 25 C, <= 100 uA at 125 C (VDS 20 V), about 16 uA at 85 C; HD_EN before Betaflight drives PIN01; <= 0.22 V vs VGS(th) 0.4 V min at 25 C (250 uA), about 0.25 V at 85 C (-2.5 mV/K); a hot low-Vin part may keep the VTX off until PIN01 is driven low (V9 hot-soak boot check). COUT on +3V3_VTX: C108 10u + C121 1u + C122 4.7u + 3x 100n + 2x 100p = 16.0 uF nominal, about 6.8 uF effective at 3.3 V typical (5.4-8.2 uF at +/-20 %; POWER-sheet DC-bias figures). Inside the LP5912 0.7-10 uF window, C108 sits at the RTC6705 VCO pins 39/40 (VCO bulk). Load 0.10 A typical, about 0.17 A at the top PA step (root power table).

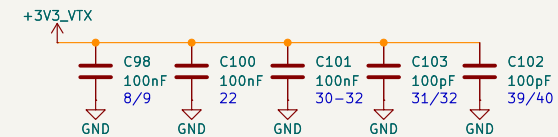
PA REFERENCE LP5907-2.85



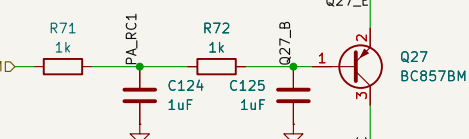
U22 LP5907-2.85 from +3V3_VTX (TI SNV5798); HD mode removes PA_VREF with the rail. EN = PA_EN straight from ESP32 GPIO2 (strap: internal 45k pull-down + the LP5907 1 MOhm EN pull-down). C123 4.7 uF 0402 (> 0.7 uF effective); CIN = C122 4.7 uF 0402 X5R at U22 IN (about 2.6 uF at 3.3 V, >= 2 uF at -20 % and hot); LP5907 DS 5.6 / 7.2.2.4 needs > 0.7 uF effective over all conditions within 1 cm; the 1 uF 0201 (about 0.45 uF) alone missed it (PA critique 3). DEVIATIONS, bench-gated: (a) VREF window: 2.85 V +/- 2 % = 2.79-2.91 V vs SE5004L VREF H 2.80-2.90 V, and the +/- 2 % is specified only for VIN >= VOUT + 1 V (VIN here 3.3 V); V5 measures PA_VREF on several units, cold and hot. (b) HD switch-on with ELRS holding PALEN high: +3V3_VTX collapses before C123 discharges, so VOUT exceeds VIN + 0.3 V (LP5907 abs max, DS 5.1) by up to one body-diode drop (<= 0.7 V, < 1 ms, <= 10 uJ from C123); V9 scopes PA_VREF and +3V3_VTX (< 0.5 V). Fallbacks: C123 down to 1 uF, or the patched ELRS drops PALEN on HD_EN.

RTC6705 VTX: 8 MHz CRYSTAL, SPI, VIDEO, PLL, PA DRIVE, PAOUT1 FEED

Decoupling at the RTC6705 supply pins (pin numbers next to each capacitor):



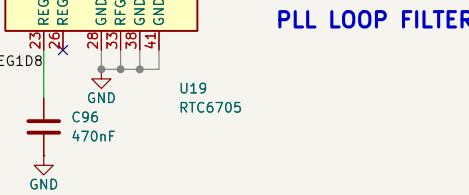
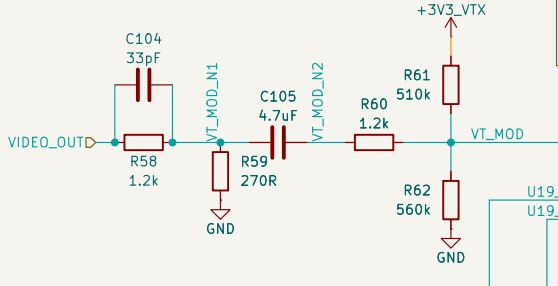
PNP PA DRIVE



PAOUT1 FEED



VIDEO MODULATION NETWORK



PLL LOOP FILTER

RTC6705, checked pin by pin against OpenOSD-X v1.01: 23 REG108.1 -> C96 470n (= ref); 26 REG108 open (ref DNI); 17 AVDD_6.5 open (ref 3.3k + 1u, no 3.3 V; V5 checks for a 6.5 MHz spur); 31/32 BUVP00/PAVDD -> +3V3_VTX with C103 100p + C101 100n shared with 30: 40 LDD0V5 -> +3V3_VTX direct with C102 100p (ref 51R + 1u filter; V5 video SNR); 39 VCOVDD with C102 + the 10 uF C108 placed there (ref 4.7u); 8/9 C98, 22 C100, Open: 1 SP1SE (internal pull-up = SPI mode), 2-4, 11-16 (11 RF_VT2 audio input; V5 spur check), 18-21, 34, 36, 37, X4 exactly 8 MHz (ELRS R = 400), CL 10 pF = 15p + 15p in series + about 2.5 pF stray.

VTX SPI (PINMAP F21), R80 / R81 / R82 1k in series with CS / CLK / DATA, drawn here and PLACED at the ESP32 END of each line (bom_plan); they limit back-feed into the unpowered RTC6705 IN HD mode (<= 2.7 mA per high line; CS idles high -> +3V3_VTX about 0.27 V on the LP5912 100 Ohm active discharge; V9 limit 0.5 V, writes only at frequency changes), ELRS HSPI 1 MHz, mode 0, LSB first; tau = 1k x about 12 pF = 12 ns (26 ns 10-90 %) against >= 20 ns setup / hold in a 500 ns half period; the same RC on all three lines, SPIDATA is bidirectional (3-wire reads through R82, production test).

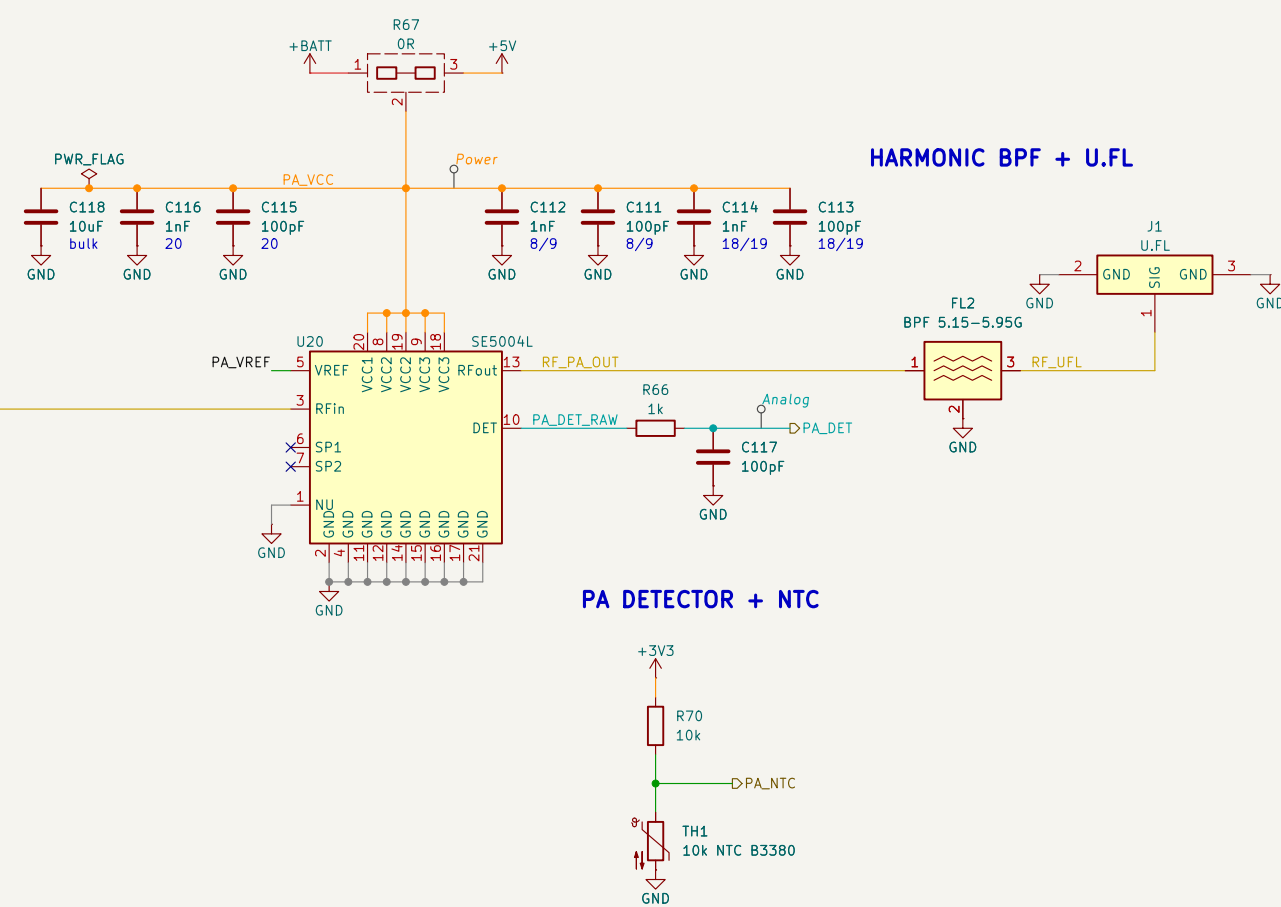
VIDEO + LOOP FILTER (OpenOSD-X v1.01), VIDEO_OUT (OSD switch: 75R camera termination on the OSD sheet) -> R58 1.2k || C104 33p -> R59 270R shunt -> C105 4.7u -> R60 1.2k -> VT_MOD, bias R61 510k / R62 560k from +3V3_VTX (1.73 V) = OpenOSD-X R24 C17 R25 C21 R29 R30 R31. Load seen at VIDEO_OUT: 1.47 kOhm DC (R58 + R59), about 0.95 kOhm at 5 MHz and 0.60 kOhm at 10 MHz (R58 / C104 4.0 MHz, R59-loaded 21.9 MHz; 270R only above about 20 MHz, outside the video band); the OSD level note uses it. Loop filter = OpenOSD-X CP -> R63 100k -> VT_VT -> C106 1u and R64 47k + C107 4.7u to GND. RTC_CP / RTC_VT / RTC_LF and VT_MOD* are high-impedance nodes (Analog net class): short copper, >= 5 mm from the boost and the ESC phases.

PNP PA DRIVE, ESP32 GPIO13 VTX AMP_PWM (ELRS vtx_amp_pwm, 10 kHz) -> two RC poles (R71 / C124, R72 / C125, 1k / 1u each) -> BC857BM base: emitter R73 (Re) to +3V3_VTX; collector Q27.C -> R65 10R -> PAOUT1_SUP -> L3 -> RTC6705 PAOUT1; it sets the RTC6705 output-stage current, not the SE5004L bias, inverting, current mode, ngspice (Nexperia BC857BM model; PAOUT1 sink model): Re 10R gives 32-50 mA at the top step over RFE 220-475 and -20.85 C, Q27 <= 114 mW (Ptot 130 mW at 85 C), base ripple 2.1 mV p-p, settles in 8 ms (ELRS 20 ms loop), GPIO13 after reset is an input with the internal pull-down (ESP32 DS IO_MUX: oe=0, ie=1, wpd: about 45 kOhm); ngspice case 4 gives 10-19 mA (Re 10R) from reset until ELRS initialize() sets the pit count (stock vtxSPIPWM = MAX_PWM), and for as long as the ESP32 stays in ROM download mode (SW1 held, recovery); Q27 about 50 mW, the PA off (GPIO2 wpd), V5 measures that boot leakage at the U.FL HD mode: both junctions reverse-biased 3.3 V (VEB0 5 V). Steps per 6 dB at Re 10R: 11-13 at 1 mA, 16-19 at 2 mA, 25-32 at 4 mA; V8 measures the PAOUT1 current at 25 mW and V5 sets Re (27R gives >= 20 steps from 1.5 mA). Start value R73 = 10R.

RATINGS AT THE TOP STEP (Re 10R, hFE 475, 85 C; 58.5 mA): R73 and R65 dissipate about 34 mW each, against the RC0201 derating (50 mW to 70 C, zero at 125 C; 36 mW at 85 C; 27 mW at 95 C; Q27 <= 114 mW against 130 mW at 85 C (12 % margin), V8 pass: top-step current and the board temperature at R73 / R65 / Q27 keep each part inside its derated rating; levers: cap the ELRS power-table top count so Ic <= 45 mA (P <= 20 mW), or Re 27R (26 mW in R73, 10 mW in R65) if V5 accepts the steps; 0402s only if V8 shows the top step is needed.

PAOUT1 FEED + DC BLOCK (OpenOSD-X L4 C40 C43 R34, fed by the PNP stage instead of its NPN follower), PAOUT1 (pin 35, +2 dBm) -> L3 4.7 nH choke to PAOUT1_SUP (C109 10 nF X7R) <- R65 10R <- Q27.C; C110 10 pF DC block into RF_PA_IN (SE5004L pin 3 'DC block required'). Only L3, C109 and C110 sit in the 5.8 GHz part-free band between pin 35 and the PA (50 Ohm CPWG < 5 mm, 45 degree). RF class on RF_PAOUT1 / RF_PA_IN / RF_PA_OUT / RF_UFL (pattern */RF_*).

SE5004L PA + SUPPLY, DETECTOR, BPF + U.FL



SE5004L PA (Skyworks 2023938 + EK1), RFout has an internal DC shunt to GND and only RFIn is marked 'DC block required'; the BPF, U.FL and antenna carry no DC, so RFout -> FL2 directly (internal output match, no match parts on the line; P5 EM-checks the line), PA_VCC = R67 selector: one 0402 OR from COM to +BATT (default, pads 1-2) or +5V (pads 2-3, hand work); the footprint anchor is the pad 1-2 midpoint, so the CPL centres the default part; VCC1/VCC2 <= VCC3 holds (one net), EK1 decoupling: 1n + 100p at 8/9, at 18/19 and at 20, on both package sides, + C118 10 uF shared, no ferrite; VREF from PA_VREF (abs max 3.6 V), SP1/SP2 (optional EVM caps) open; NU (open in the package) tied to GND; it lies in the DS ground metal, EP + pins 2, 4, 11, 12, 14-17 to GND with a via field.

PA DETECTOR + NTC (spec 4.8), DET (0.325 V no RF, 1.0 V at 27 dBm) -> PA_DET_RAW -> R66 1k / C117 100p -> PA_DET (ESP32 GPIO4, ADC2); the trace crosses sides, keep the RC at the ESP32 end of R66, TH1 NCP03XH103F05RL in the PA ring + R70 10k from +3V3 -> PA_NTC (GPIO34, ADC1, input only); 1.65 V at 25 C, BPF + U.FL: FL2 Walsin RFBPF1608060K98Q1C on the PA side away from the PA body (power handling at 0.71 W; open item 07); J1 U.FL with L2 cut out under the signal pad (RF_PAD_CUTOUT), >= 4 GND vias at the tabs.

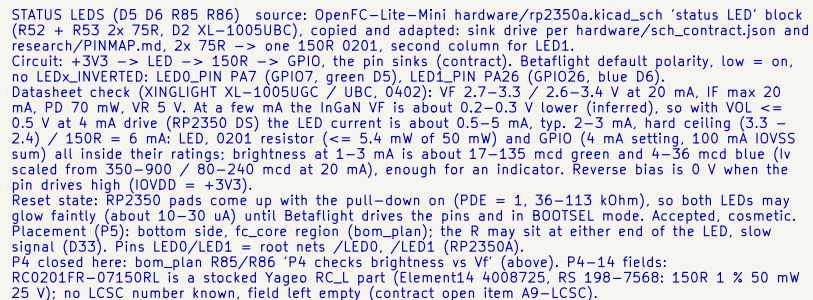
SHEET NOTES

SHEET SOURCE: Generated by gen_vtx.py (hardware/tools/sch_build.py + sch_root.py). No OpenDrone sibling sheet or commons block exists for an RTC6705 VTX, so the circuit follows the OpenOSD-X v1.01 reference schematic (RTC6705 supply pins, video network, loop filter, PAOUT1 feed), the Skyworks SE5004L datasheet 2023938 + EK1 and DESIGN-SPEC 4.8. Refs, values, MPNs and footprints: hardware/bom_plan.json. Signal flow: ESP32 controls (RX sheet) on the left, RTC6705 in the middle, PA, BPF and U.FL on the right.

BENCH ITEMS on this sheet: RTC6705 pin 11 RF_VT2 open (V5 spur check); PA drive Re final value (V5 / V8, 27R fallback line); PA_VREF window and the HD switch-on excursion (V5 / V8; PA REFERENCE note); VIDEO_OUT OSD contrast and edges (V9, OSD sheet); X4 start-up margin (V5: series R 3-5x the 250R ESR, -10 C cold start, slow supply ramp); Q27, R73, R65 temperature at the top step (V5 / V8, PNP PA DRIVE note).

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Sheet: /VTX/ File: vtx.kicad_sch		
Title: 5.8 GHz VTX: RTC6705 + SE5004L PA		
Size: A2	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6	Id: 12/14	



OpenDrone

Rev: rev1
Id: 13/14

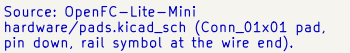
Battery pads (15)

Source: OpenFC-Lite-Mini hardware/pads.kicad_sch (Conn_01x01 pad, pin down, rail symbol at the wire end).

J2 '+' = +BATT_IN, J3 '-' = GND: BT2.0 pigtail (22 AWG, user-fitted) through plated D1.1 holes in 3.0 x 2.0 mm pads (lib:BatPad_3.0x2.0mm_PTH1.1), >= 12 power vias each into the pours (spec 4.10, D72 power plan: +BATT_IN into L4 + L1/L6, GND into L2/L5). Silk '+' / '-' 2.0 mm and '1S' (spec 13).

1S LiPo/LiHV only, 3.0–4.35 V (D14). D1 SMF5.0A TVS, the pad bulk and the shunt R1 sit on POWER, which also holds the +BATT_IN and GND PWR_FLAGS. Netclass VBAT: directive here on +BATT_IN at J2.

Copper only: DNP, out of BOM and CPL (LINEUP A16).



J2 '+' = +BATT_IN, J3 '-' = GND; BT2.0 pigtail (22 AWG, user-fitted) through plated D1.1 holes in 3.0 x 2.0 mm pads (lib:BATTPad_3.0x2.0mm_PTH1.1), >= 12 power vias each into the pours (spec 4.10, D72 power plan; +BATT_IN into L4 + L1/L6, GND into L2/L5). Silk '+' / '-' 2.0 mm and '1S' (spec 13).

1S LiPo/LiHV only, 3.0–4.35 V (D14). D1
SMF5.0A TVS, the pad bulk and the shunt R1
sit on POWER, which also holds the +BATT_IN
and GND PWR_FLAGS. Netclass VBAT: directive
here on +BATT_IN at J2.

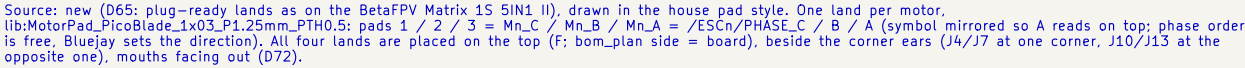
Copper only: DNP, out of BOM and CPL (LINEUP A16).

Motor lands M1–M4: plug-ready, Molex PicoBlade header DNP

Source: new (D65: plug-ready lands as on the BetaFPV Matrix 1S 5IN1 II), drawn in the house pad style. One land per motor, lib:MotorPad_PicoBlade_1x03_P4.25mm_PTH0.5; pads 1 / 2 / 3 = Mn_C / Mn_B / Mn_A = /ESCn/PHASE_C / B / A (symbol mirrored so A reads on top; phase order is free, Bluejay sets the direction). All four lands are placed on the top (F; bom_plan side = board), beside the corner ears (J4/J7 at one corner, J10/J13 at the opposite one), mouths facing out (D72).

Motor wires solder to the land (pads 1/3 1.65 x 1.50, pad 2 0.90 x 1.50) or through the 0.50 mm plated holes: the recommended default. The Molex 53047–0310 PicoBlade header (board side of the hobby 'JST1.25' motor plug, wire side 51021–0300) is a DNP BOM line, out of the CPL, hand-fitted from the far side with its pins trimmed on the land side. Molex rates 1.0 A per contact, −40..85 C, far below the motor current (4.3–5.2 A per channel, 12 A bursts): the published ESC ratings apply to soldered motor wires; the README pinout states the 1.0 A contact rating and that plugged motors run outside it at the user's risk (BetaFPV Matrix practice); V4 records the header temperature. Mating zone (D72): the plug faces outward over the ear neck, clear of parts.

Net names come from the ESC sheets (the root wires carry no label: /ESCn/PHASE_x). Netclass Phase: directive on M1_A here, the pattern /ESC?/PHASE_? assigns the rest. Phase copper runs from the AGM210MAP phase island straight into the land; solder areas 0.30 mm and phase copper 0.20 mm from the edge (SOLDER_PAD class). M1 rear right (ESC1), M2 front right (ESC2), M3 rear left (ESC3), M4 front left (ESC4); Betaflight quad–X.



Motor wires solder to the land (pads 1.3 x 1.65 x 1.50, pad 2.090 x 1.50) or through the 0.50 mm plated holes; the recommended default, The Molex 53047-0310 Picoblade header (board side of the hobby "JST1.25" motor plug, wire side 51021-0300) is a DNP BOM line, out of the CPL, hand-fitted from the far side with its pins trimmed on the land side. Molex rates 1.0 A per contact, -40..85 C, far below the motor current (4.3-5.2 A per channel, 12 A bursts); the published ESC rating apply to soldered motor wires; the README pinout states the 1.0 A contact rating and that plugged motors run outside it at the user's risk (BetaFPV Matrix practice); V4 records the header temperature. Mating zone (D72): the plug faces outward over the ear neck, clear of parts.

Net names come from the ESC sheets (the foot wires carry no label: /ESCn/PHASE_x). Netclass Phase: directive on M1_A here, the pattern /ESC?/PHASE_? assigns the rest. Phase copper runs from the AGM210MAP phase island straight into the land; solder areas 0.30 mm and phase copper 0.20 mm from the edge (SOLDER_PAD class). M1 rear right (ESC1), M2 front right (ESC2), M3 rear left (ESC3), M4 front left (ESC4): Betaflight quad-X.

USB: JST SH 1.0 mm 4-pin + ESD

Source: OpenFC-Lite hardware/pads.kicad_sch (JST SH SM04B plug in a frame; there it carries a UART). The USB pinout, D7 and the VBUS flag are new (D57). Commons block USB_C_GCT4105_USBL6 checked, not used (USB-C and a 6-pin ESD part).

J31 JST BM04B-SRSS-TB(LF)(SM) as in bom_plan (vertical, bottom): 1 GND, 2 D- (USB_D_N), 3 D+ (USB_D_P), 4 VBUS, tabs 5/6 GND = BetaFPV 'USB(GND DM DP 5V)' order (bom_plan, IO-SET row 6, LIBRARY D57). Pin 1 = GND is inferred: V-plan gate V0-USB before the P5 freeze (continuity on a BetaFPV SH1.0 USB adapter or a Matrix II close-up); a mirrored plug would put 5 V on board GND and -5 V on the SGM40661 IN (abs min -0.3 V). D76 (owner) keeps the vertical BM04B facing up next to a mounting hole (D72 side-entry SM04B withdrawn); its mating zone above the board stays clear of tall parts.

D7 TPD2EUSB30DRTR (TI SLVSAC2G): pin 1 D+, 2 D-, 3 GND (Table 5-1); VRWM 5.5 V, IO abs max 6 V, so a D+/D- short to VBUS survives; 8 V clamp at 1 A, 0.7 pF, IEC 61000-4-2 8 kV contact, -40..85 C. It sits at J31, ahead of the 27 R R42/R43 at the RP2354A (RP2350A sheet). Layout per D5 10.1: next to J31, protected traces straight with rounded corners, no unprotected trace beside the J31-D7 stretch, GND pin by via-in-pad into L2. USB pair 90 ohm, length-matched (D33); netclass USB from the directive on USB_D_P (P/_N pair).

VBUS is a hierarchical net (LINEUP A11) to POWER: U24 SGM40661 OVP IN + C131 100 nF take the hot-plug and over-voltage duty (D57). PWR_FLAG here: J31 is the only source of VBUS. A cable with pins 1/4 swapped puts -5 V on VBUS (unprotected, as on the Matrix).



J31 JST BM04B-SR5S-TB(LF)(SN) as in bom-plan (vertical, bottom): 1 GND, 2 D- (USB-D_N), 3 D+ (USB-D_P), 4 VBUS, tabs 5/6 GND = BetaFPV "USB(GND DM DP SV)" order (bom_plan, IO-SET row 6, LIBRARY D57). Pin 1 = GND is inferred: V-plan gate V0-USB before the P5 freeze (continuation on a BetaFPV SH1.0 USB adapter or a Matrix II close-up); a mirrored plug would put 5 V on board GND and -5 V on the SGM40661 IN (abs min -0.3 V). D76 (owner) keeps the vertical BM04B facing up next to a mounting hole (D72 side-entry SM04B withdrawn); its mating zone above the board stays clear of tail parts.

D7 TP02D2E5B.03.0R7 (T1 SILVACAG2): pin 1, D+ - 3, GND (Table 5-5): VRWM 5.5 V, IO abs max 6 V, so a D+/- short to VBUS survives; 8 V clamp at 1 A, 0.07 pF, IELV 61000 - 4.2 k Ω contact, -40...85 C. It sits at J31, ahead of the 27 R R42/R43 at the RP235A4 (RP2350A sheet). Layer per DS 10.1: next to J31, protected traces straight with rounded corners, no unprotected trace beside the J31-D7 stretch, GND pin by via-in-pad into L2. USB pair 90 ohm, length-matched (D33): netclass USB from the directive on USB_D_P ($P_{-}N$ pair).

VBUS is a hierarchical net (LINEUP A11) to POWER: U24 SGM40661 OVP IN + C131 100 nF take the hot-plug and over-voltage duty (D57). PWR_FLAG here: J31 is the only source of VBUS. A cable with pins 1/4 swapped puts -5 V on VBUS (unprotected, as on the Matrix).

User UART pads (PIOUART0)

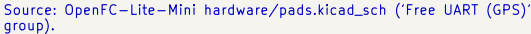
Source: OpenFC-Lite-Mini hardware/pads.kicad_sch ("Free UART (GPS)" group).

TP0 = PIOUART0_TX (GPIO2), RP0 = PIOUART0_RX (GPIO3), PIO1 shared with the LED strip (PINMAP F5), half duplex and inversion capable. 8N1 only; Betaflight's RP2350 PIO UART rejects even parity, so 5BUS / 8E2 receivers use the EXT—USER—PADS preset (hardware UART0 on these pads; 8E2 needs a Betaflight release with the uart_hw.c fix); I2C1 via I2C—USER (IO—SET 2.3).

5V = +5V (TPS2116 output, 0.05 A allowance, spec 5.1) and GND for the device.

P4–11 (pads, power—up): GPIO2/3 are FT pads, 5.5 V tolerant with IOVDD up but limited to 3.63 V while IOVDD is still 0 (PINMAP F16). The 5V pad comes up before +3V3 (LDO start after the mux): use 3.3 V—logic devices (GPS, receivers); a 5 V—logic TX on RP0 can exceed that limit during the ramp.

Silk TP0 RP0 5V GND; copper only (DNP, out of BOM and CPL, LINEUP A16).



P0 = PIOUART0_TX (GPIO2), R0 = PIOUART0_RX (GPIO3), P101 shared with the LED strip (PINMAP F5), half duplex and inversion capable. 8N1 only; Betaflight's RP2350 PIO UART rejects even parity, so SBUS / BE2 receivers use the EXT-USER-PADS preset (hardware UART0 on these pins). BE2 needs a Betaflight release with the uart_hw.c fix; I2C1 via I2C-U (10-SET 2.3).

5V = +5V (TPS2116 output, 0.05 A allowance, spec 5.1) and GND for device.

P4-11 (pads, power-up): GPIO2/3 are FT pads, 5.5 V tolerant with IOVDD up but limited to 3.63 V while IOVDD is still 0 (PINMAP F16). The 5V pad comes up before +3V3 (LDO start after the mux): use 3.3 V–logic devices (GPS, receivers); a 5 V–logic TX on RP0 can exceed that limit during ramp.

Silk TP0 RPO 5V GND; copper only (DNP, out of BOM and CPL, LINEUP A16).

LED strip + buzzer row

Source: OpenFC-Lite-Mini hardware/pads.kicad_sch (LED-strip and buzzer pads).

Row GND LED 5V BZ- (D63/D64): the strip gets its own GND pad J33 (supersedes D27).

LED = LED_STRIP driven straight from GPIO8 (PI01 ws2812; D52 BR-04; no translator, no series resistor, Matrix II parity). A WS2812B on 5 V asks VIH 0.7 VDD = 3.5 V; the 3.3 V drive works on the common whoop strips, a level problem shows as a dead first LED (bench check, V9).

5V = +5V for the strip (0.20 A allowance, analog mode, spec 5.1) and the buzzer +. BZ- = BUZZER-; drain of the AP1606 low-side switch Q26, gate on GPIO17 (BEEPER-INVERTED, RP2350A sheet); the buzzer sits between 5V and BZ-.

Copper only (DNP, out of BOM and CPL, LINEUP A16).



Row GND LED 5V BZ- (D63/D64): the strip gets its own GND pad J33 (supersedes D27).

LED = LED_STRIP driven straight from GPIO8 (PIO1 ws2812; D52 BR=0, no translator, no series resistor, Matrix II parity). A WS2812B on 5 V and VIH 0.7 VDD = 3.5 V; the 3.3 V drive works on the common whoop strip; a level problem shows as a dead first LED (bench check, V9).

5V = +5V for the strip (0.20 A allowance, analog mode, spec 5.1) and the buzzer +. BZ- = BUZZER-; drain of the AP1606 low-side switch gate on GPIO17 (BEEPER_INVERTED, RP2350A sheet); the buzzer sits between 5V and BZ-.

Copper only (DNP, out of BOM and CPL, LINEUP A16).

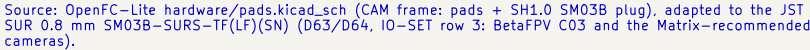
Camera pads + JST SUR 0.8 mm plug

Source: OpenFC-Lite hardware/pads.kicad_sch (CAM frame: pads + SH1.0 SM03B plug), adapted to the JST SUR 0.8 mm SM03B-SURS-TF(LF)(SN) (D63/D64, IO-SET row 3: BetaFPV C03 and the Matrix-recommended cameras).

Plug and pads in parallel: J32 pin 1 VIDEO_IN, 2 GND, 3 +5V_CAM, tabs 4/5 GND: pads CAM / GND / 5V (J23 J25 J24). On the board the plug faces inward and the pads sit in front of its mouth: plug OR solder in one mating zone (D72).

VIDEO_IN -> OSD front end (TLV7031 sync + SN74LVC1G3157, OSD sheet), netclass Analog (OSD directive). +5V_CAM = +5V after FB1 120 R + C15 10 uF (POWER), 0.12 A. Video return: CAM GND pad and plug pin 2 enter L2 at the CAM pad (spec 11).

Gate G1 still open: pin order provisional until calipers + photo on a C03 lead and the MX2 CAM IN. With GND in the middle a mirrored footprint puts 5 V on the camera's video output. G1 = 1.0 mm pitch -> SM03B-SR5S-TB(LF)(SN), same pins.



Plug and pads in parallel: J32 pin 1 VIDEO_IN, 2 GND, 3 +5V_CAM, tabs 4/5 GND; pads CAM / GND / 5V (J23 J25 J24). On the board the plug faces inward and the pads sit in front of its mouth: plug OR solder in one mating zone (D72).

VIDEO_IN -> OSD front en (TLV7031 sync + SN74LVC1G3157, OSD sheet), netclass Analog (OSD directive +5V_CAM = +5V after FB1 120 R + C15 10 uF (POWER), 0.12 A. Video return: CAM GND pad and plug pin enter L2 at the CAM pad (spec 11).

Gate G1 still open: pin order provisional until calipers + photo on a C03 lead and the MX2 CAM IN. With GND in the middle a mirrored footprint puts 5 V on the camera's video output. G1 = 1.0 mm pitch -> SM03B-SRSS-TB(LF)(SN), same pins.

HD pads (DJI O4 Lite)

Source: OpenFC-Lite-Mini hardware/pads.kicad_sch (digital video pads), new pad set (D13/D16).

VHD = +5V_HD from U5 TPS22810 (on with HD_EN and cell $\geq 3.23\text{--}3.82\text{ V}$, held down to $2.74\text{--}3.14\text{ V}$; POWER P4-1, 1 % resistors), 1.20 A budget; O4_Lite input $3.7\text{--}13.2\text{ V}$. TX1 = UART1_TX (GPIO4) $\rightarrow$ O4_RX, RX1 = UART1_RX (GPIO5) $\leftarrow$ O4_TX; MSP DisplayPort; no SBUS pad (D16). Cut DJI cable: red $\rightarrow$ VHD, black + brown $\rightarrow$ GND, white $\rightarrow$ TX1, grey $\rightarrow$ RX1, yellow open (RP0 only with the EXT-USER-PADS preset).

Analog builds: TX1/RX1 are a free hardware UART, or I2C0 via the I2C-HD preset (IO-SET row 8).

P4-11: the O4 stays unpowered until HD_EN, so its TX cannot drive RX1 before IOVDD (FT pads limited to 3.63 V unpowered, PINMAP F16). A device on TX1/RX1 in an analog build runs from the 5V pads, which come up before +3V3: 3.3 V logic only (a 5 V TX on RX1 can exceed 3.63 V during the ramp, as on RP0).

Copper only (DNP, out of BOM and CPL, LINEUP A16).



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VHD = +5V_HD from U5TPS22810 (on with HD_EN and cell >= 3.23-3.82 V, held down to 2.74-3.14 V; POWER_P4 = 1, 1% resistors), 1.20 A budget; Q4 Lite input 3.7-13.2 V, TX1 = UART1_TX (GPIO4) -> Q4 RX, RX1 = UART1_RX (GPIO5) <- Q4 TX; MSP DisplayPort; no SBUS pad (D16). Cut DJI cable: red -> VHD, black + brown -> GND, white -> TX1, grey -> RX1, yellow open (RP0 only with the EXT-USER-PADS preset).
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Analog builds: TX1/RX1 are a free hardware UART, or I2C0 via the I2C-HD preset (IO-SET row 8).

P4-11: the O4 stays unpowered until HD_EN, so its TX cannot drive RX1 before IOVDD (FT pads limited to 3.63 V unpowered, PINMAP F16). A device on TX1/RX1 in an analog build runs from the 5V pads, which come up before +3V3: 3.3 V logic only (a 5 V TX on RX1 can exceed 3.63 V during the ramp, as on RP0).

Copper only (DNP, out of BOM and CPL, LINEUP A16).

Mounting holes



H1

MountingHole 3,5



H2

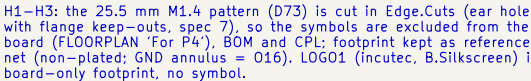
MountingHole 3,5



H3

MountingHole 3,5

H1–H3: the 25.5 mm M1.4 pattern (D73) is cut in Edge.Cuts (ear holes with flange keep-outs, spec 7), so the symbols are excluded from the board (FLOORPLAN 'For P4'). BOM and CPL: footprint kept as reference. No net (non-plated: GND annulus = 016). LOG01 (Incutech, B.Silkscreen) is a board-only footprint, no symbol.



PADS: every off-board connection. Style from the OpenFC-Lite-Mini and OpenFC-Lite pads sheets (pads pin down, a rail symbol or a hierarchical label at the wire end, plug blocks in frames); the source of each block is in its note. References, values (pad codes = silk codes), footprints, MPN / Manufacturer / LCSC and the DNP / BOM / CPL attributes are hardware/bom_plan.json; solder pads are DNP and out of BOM and CPL (LINEUP A16), the motor header is a DNP BOM line.

22 hierarchical labels = the PADS sheet pins (sch.contract,json): VIDEO_IN, M1_A..M4_C, P10UART0_TX/RX, UART1_TX/RX, LED_STRIP, BUZZER--, USB_D_P/N, VBUS, Rails by power symbol (+BATT_IN, +5V, +5V_CAM, +5V_HD, GND; sources on POWER); VBUS carries its PWR_FLAG here. Netclass directives: VBAT (+BATT_IN at J2), Phase (M1_A), USB (USB_D_P).
 (Owed: J31 pin 1 = GND (gate V0=USB before the P5 freeze), J32 pin order (gate I1), motor-header temperature (V4).

OpenDrone

Sheet: /PADS/
File: pads.kicad_sch
Title: Pads, motor lands, camera, HD, USB

Size: A3	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 14/14